DATA WAREHOUSE

The base for Business intelligence

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Definition

2

Architectures

3

DFM

4

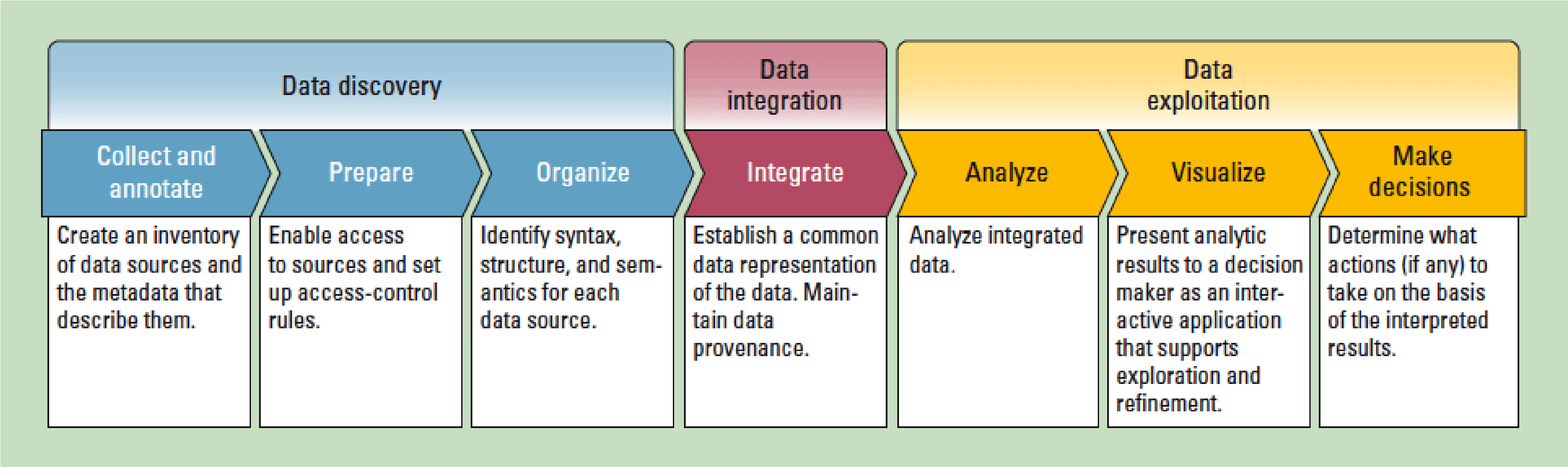
Star schema

5

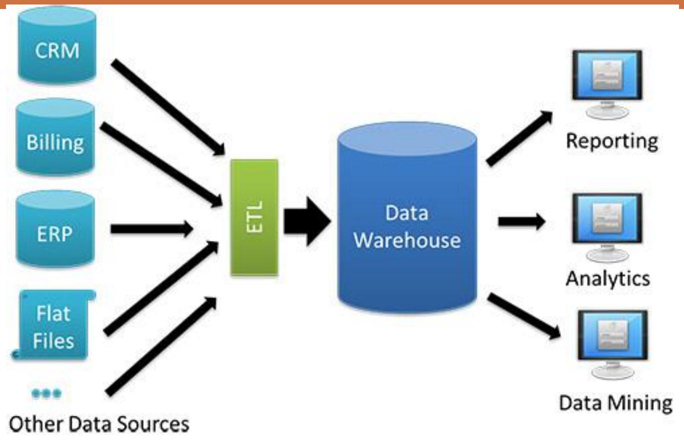
Slow changing dimension

DATA VALUE CHAIN

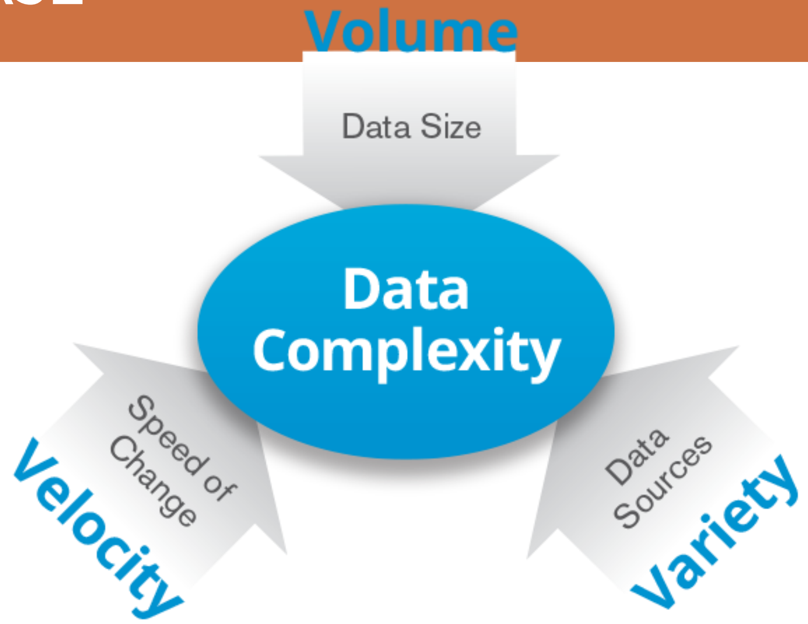
- https://www.computer.org/csdl/mags/it/2013/01/mit2013010057-abs.html



HOW TO MANAGE LARGE DATABASE

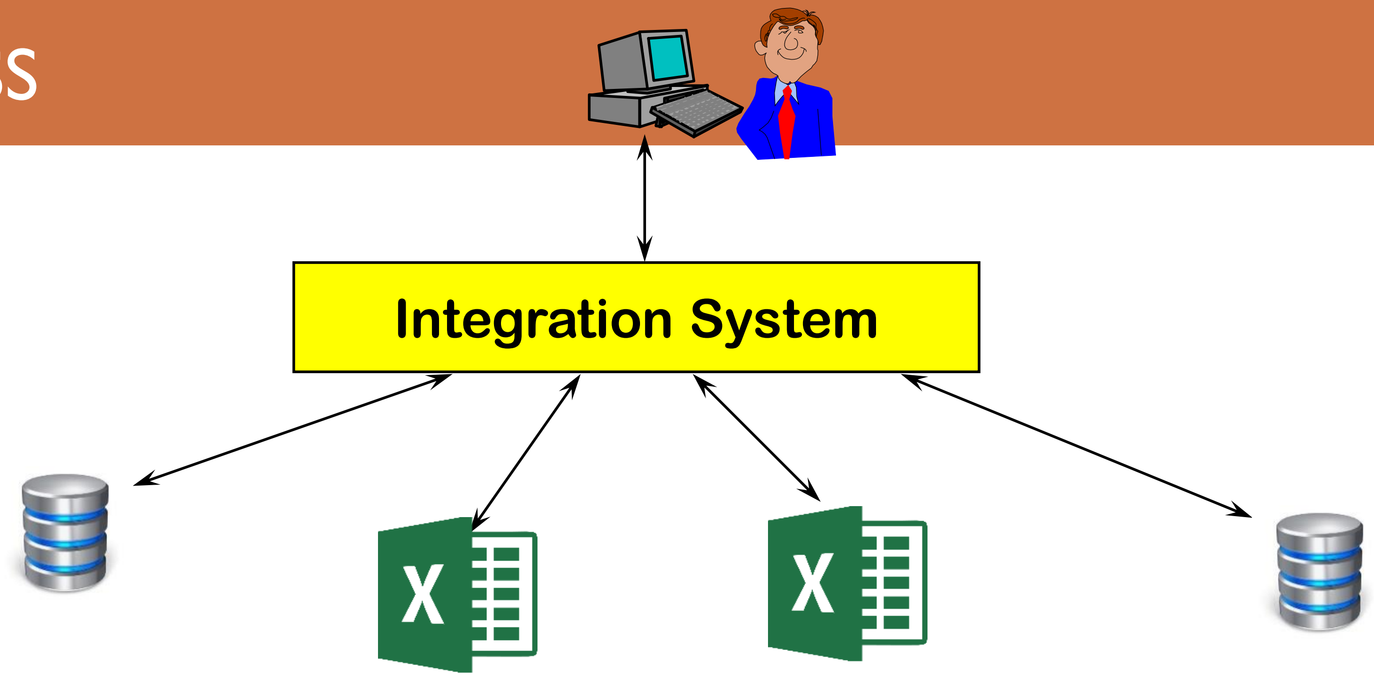


Datawarehouse



Big data

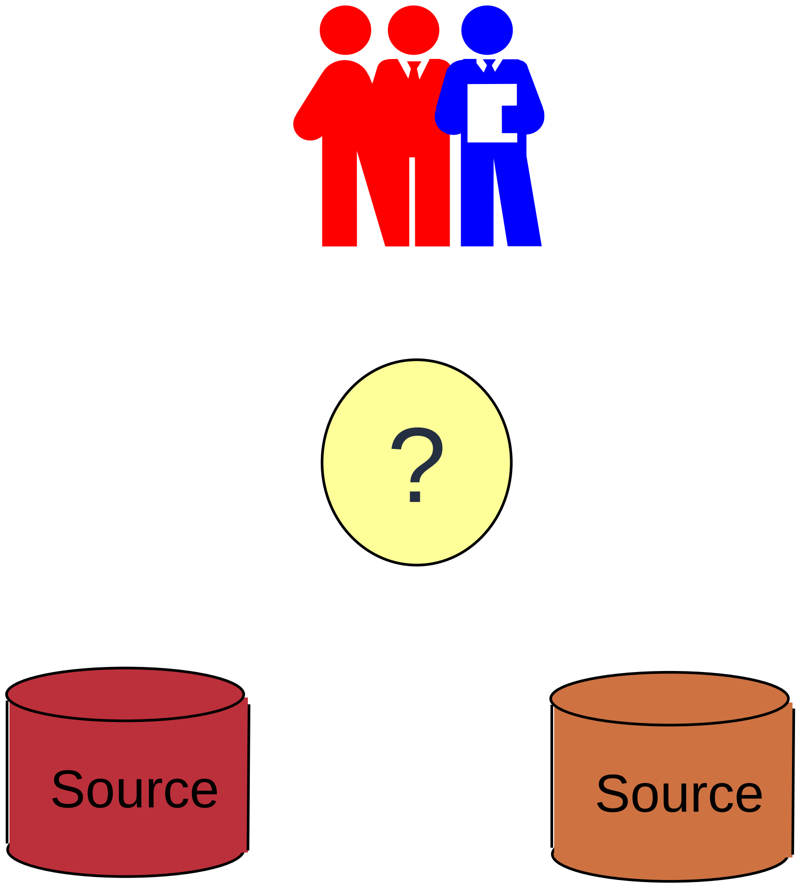
UNIFIED ACCESS



- Collect data from heterogeneous sources
- Provide an integrate and shared view of data

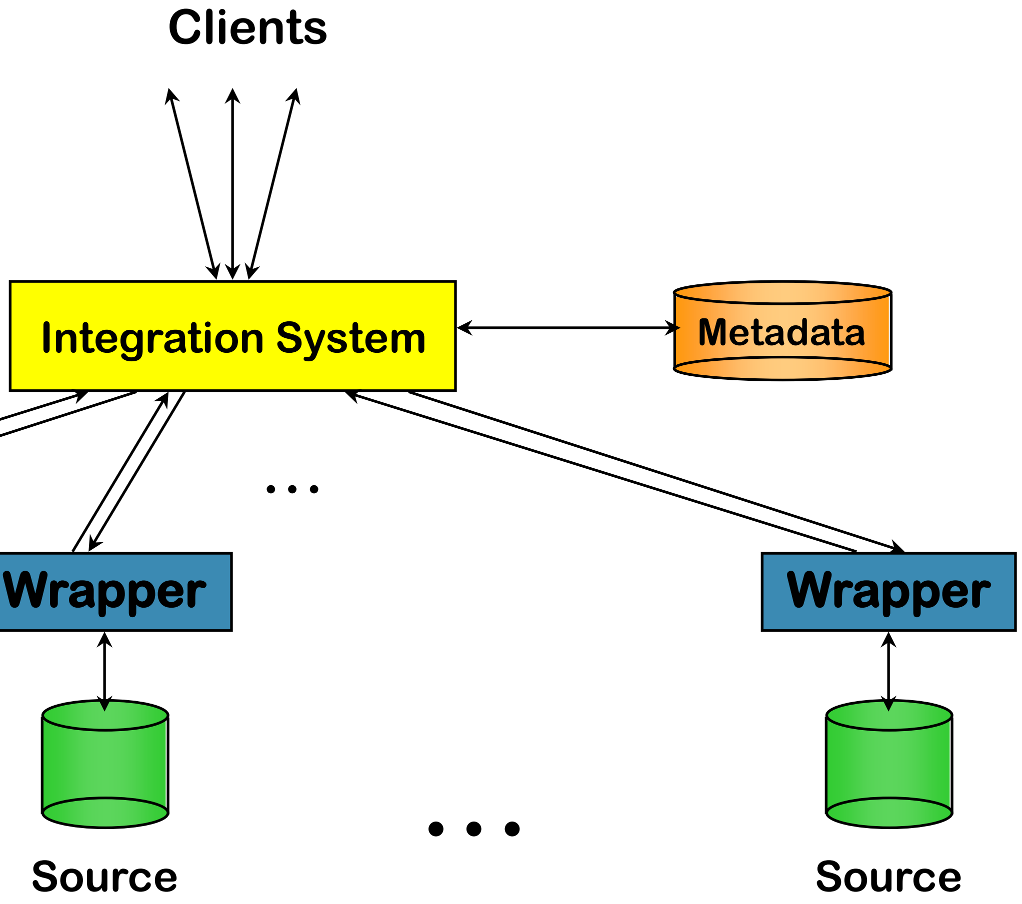
SOLUTIONS

- Two different solution
  - Data integration (Lazy)
  - Warehouse (Eager)



DATA INTEGRATION

- Query-driven (lazy, on-demand)

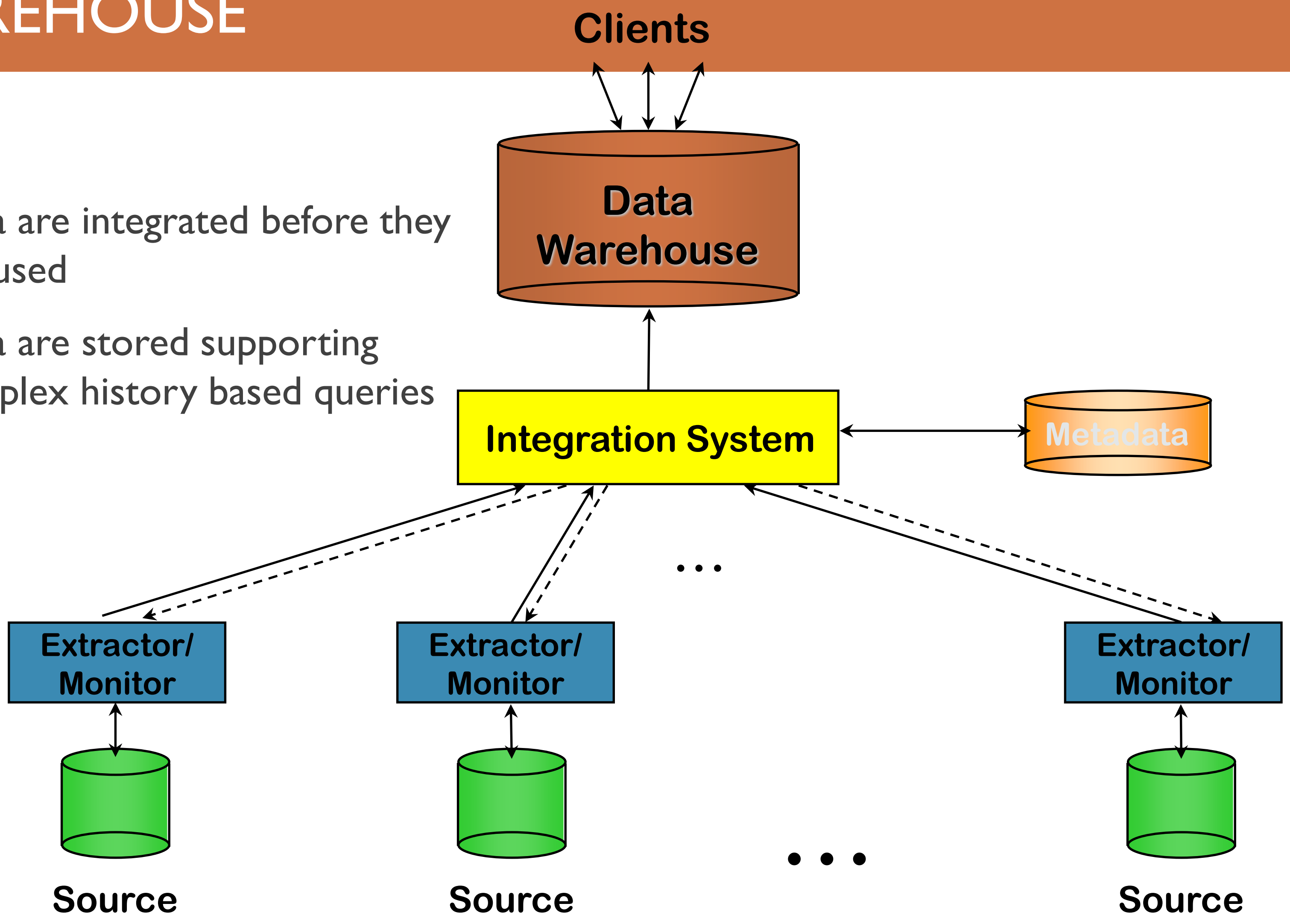


ISSUES:

- Query
  - Source not available
  - Complext on the fly integration
- Coslty and inefficient solution for complex query
- No historical data
- Increase workload

DATAWAREHOUSE

- Data are integrated before they are used
- Data are stored supporting complex history based queries



CONS

- Create the DW
  - The warehousing process could be very long and it involves multiple areas of a company
- Quality
  - Data integration allow discovering a huge data quality issues. Moreover assess and imporve data it is not an easy task
- Evolution
  - New data source to integrate or new type of queries can impose long and costly redesign of DW

PROS

- Fast complex query
  - For read only data
- No interference with operational sources
  - Complex query can impact efficiency
- Data are stored into a data warehouse
  - Data can be modified, redesigned and so on
  - Data can store historical series
  - Better access controllo
- <u>*Data warehouse is still used as first data analytic tool in a large number of companies*</u>

DW DEFINITION

A data warehouse is simply a single, complete, and consistent store of data obtained from a variety of sources and made available to end users in a way they can understand and use it in a business context."

-- Barry Devlin, *IBM Consultant*

A DW is a

- subject-oriented,
- integrated,
- time-varying,
- non-volatile

collection of data that is used primarily in organizational decision making."

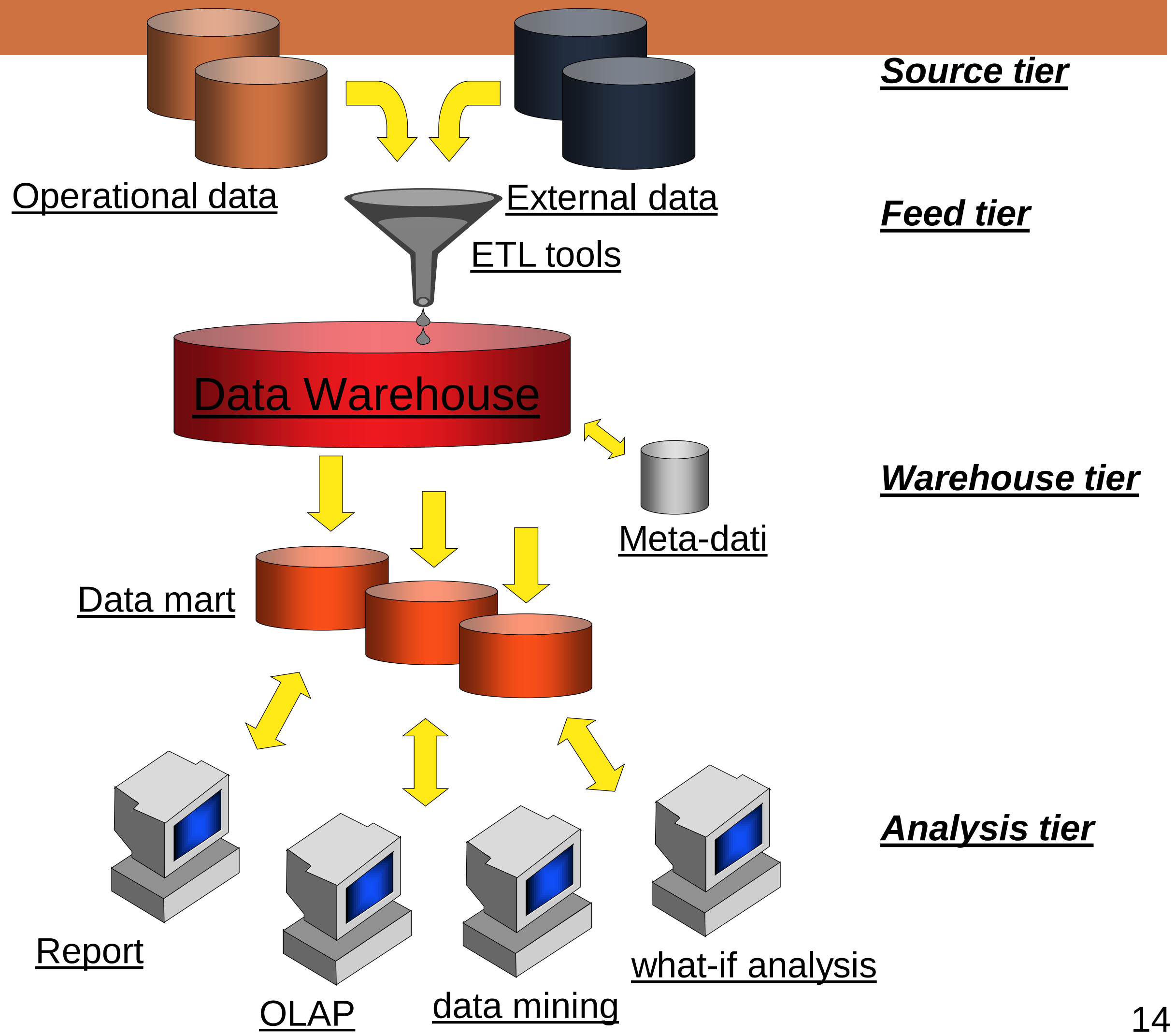
-- W.H. Inmon, Building the Data Warehouse, 1992

DATAWAREHOUSE ARCHITECTURE

TWO TIERS ARCHITECTURE

DATA MART:

It is a subset or an aggregation of data in the primary DW. Data mart includes set of data rilevant for specific business area or type of business and so on.



2 TIERS ARCHITECTURE

- Data mart derived from primary DW are named *dependent*. Such solution was adopted in large organizations:
  - For incremental development strategy;
  - DW defines very well the boundaries of relevant information for specific business needs and related queries;
  - Data mart has reduced dimensions wrt DW and so the efficiency increases
- In some case data marts are directly fed by data sources. In this case, we call them *independent*
  - Lack of primary DW makes the design process easy but the design of schema itself can be complex and in case of multiple independent data marts conflict may raise

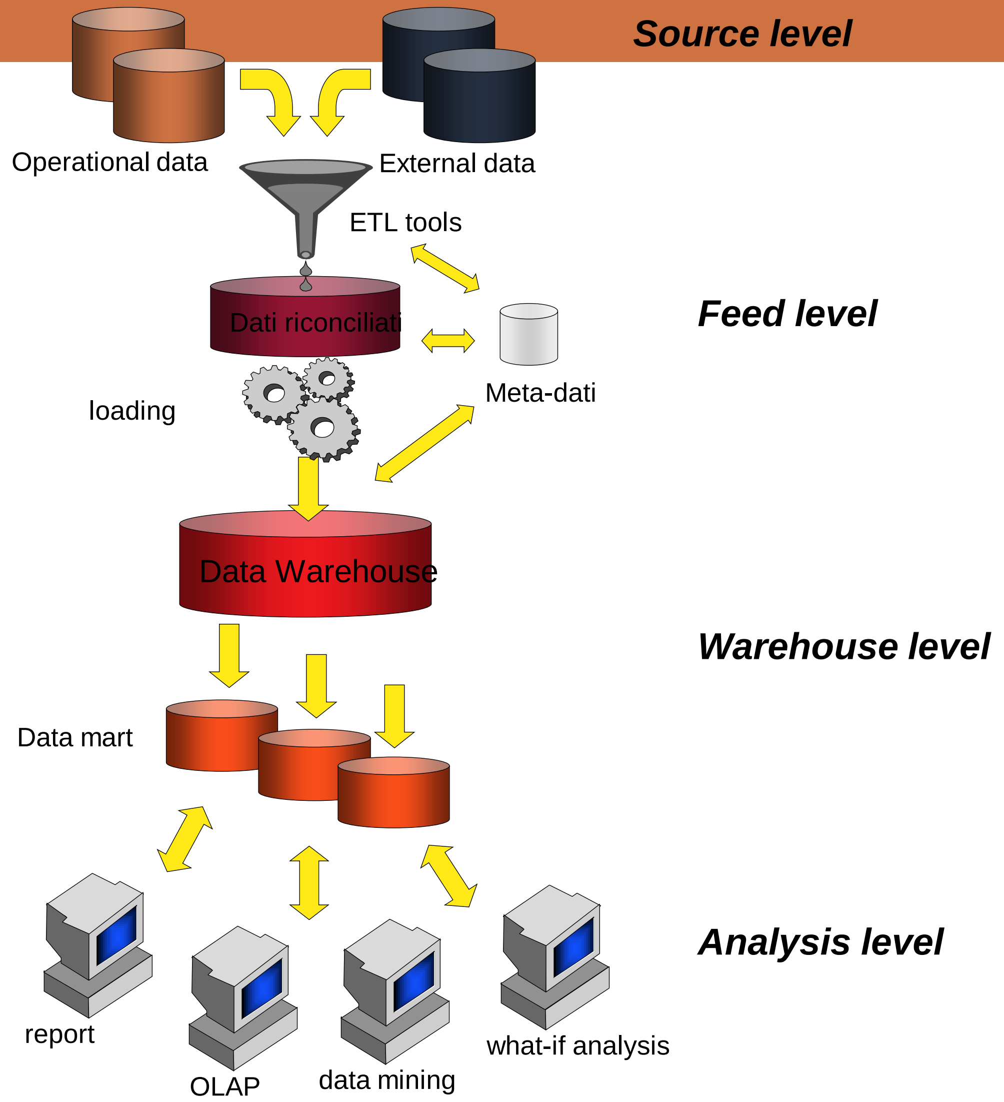
TWO TIERS

- Pros:
  - This architecture allows users to have high quality data even if it is no possibile to access original data sources
  - Complex queries does not impact data sources
  - At logical level the DW is based on a **multidimensional model**, while all data sources are defined with relational model
  - DW systems implements specific techniques for suporting large amount of data

3 TIERS ARCHITETTURE

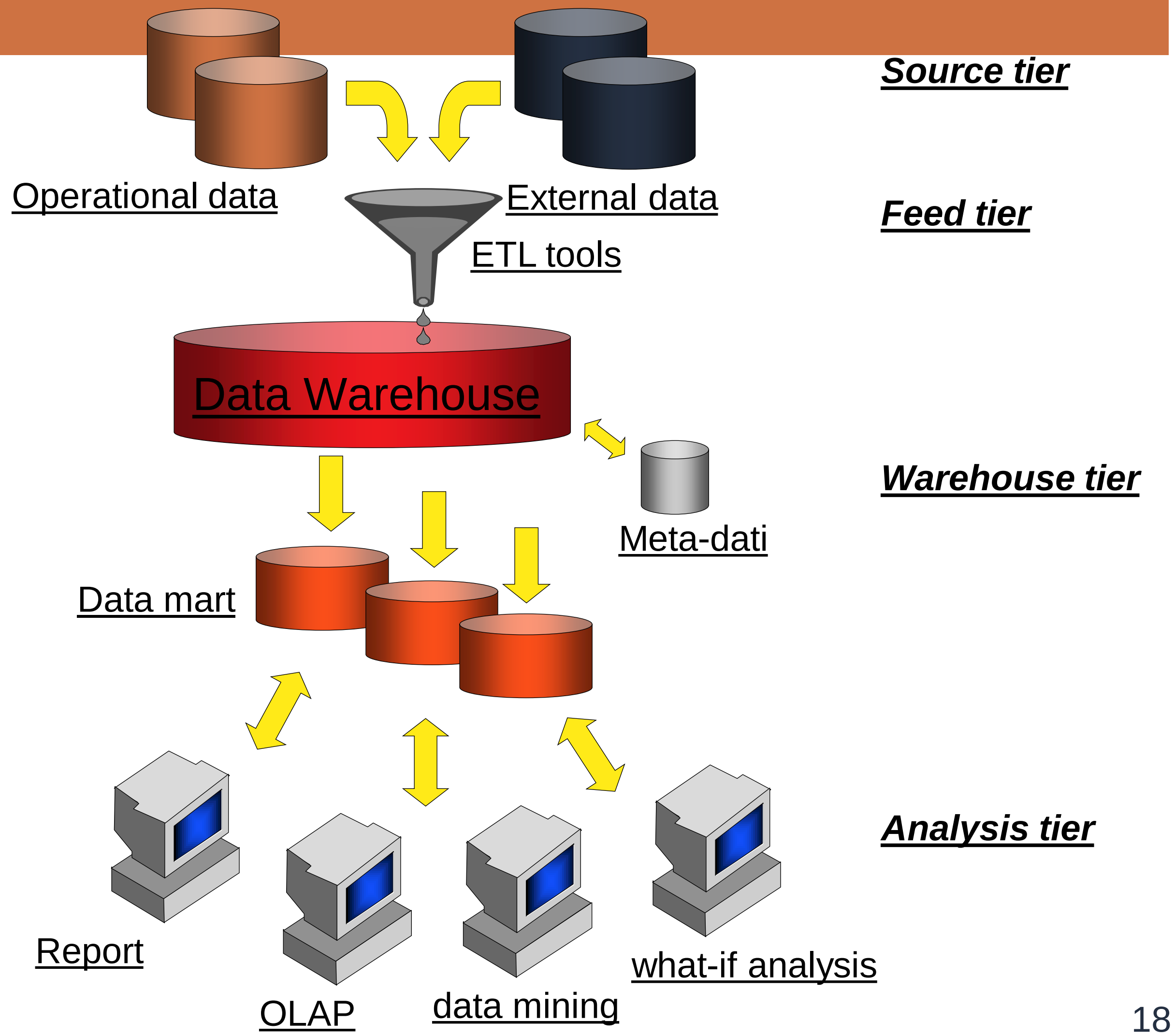
Reconciled data

Operational data created after integration and quality process of data sources: data are integrated, coerent, correct, up-to-date, and detailed



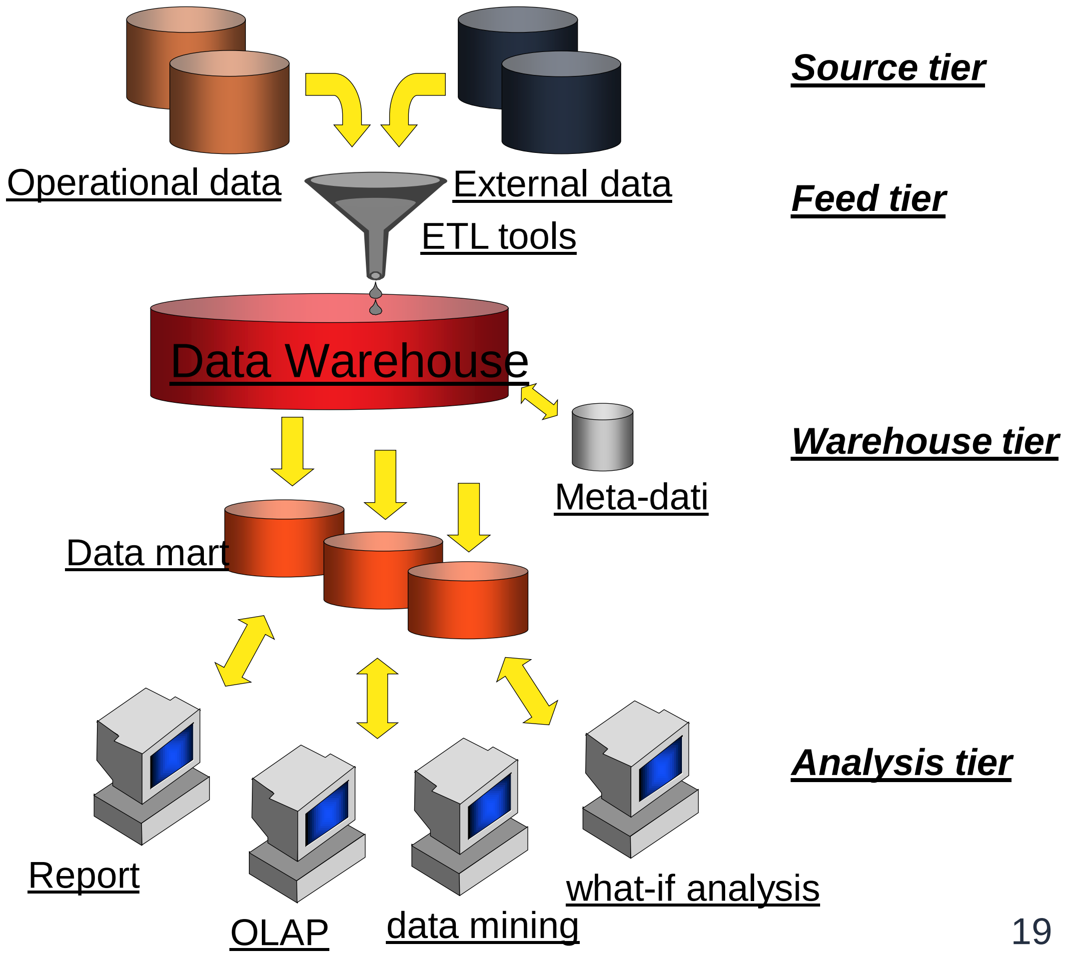
HUB-AND-SPOKE

One of the most used architecture in large company



CENTRALIZED DATA WAREHOUSE

- Proposed by Inmon
- More a logical approach than a real one

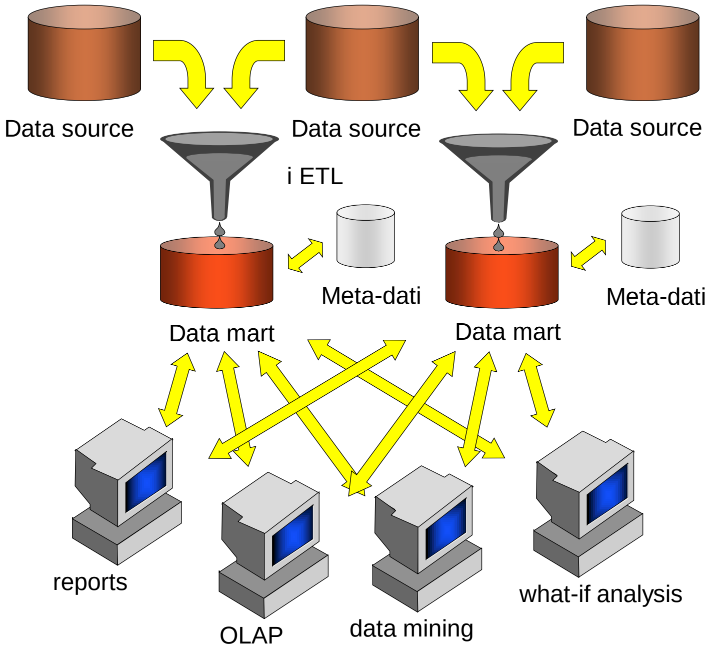


2 TIERS ARCHITECTURE

- Main advantage is the reconcilied data layer.
- Unique data model is created for all company
- Clear separation between extration pahse and the feed phase

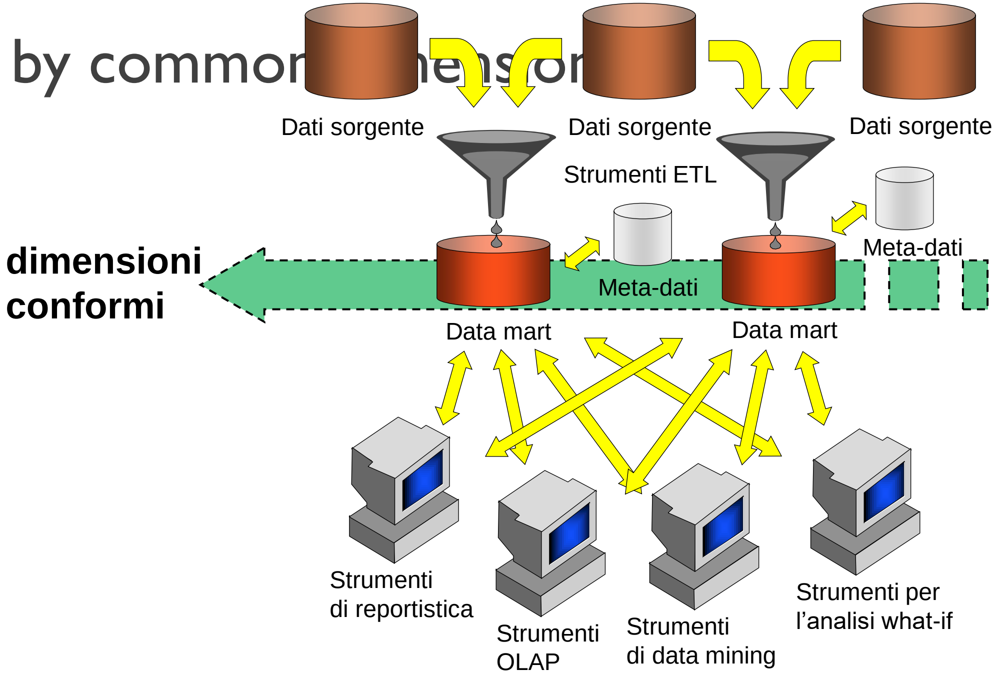
INDEPENDENT DATA MART

- Each organizational unit develop his dm independent
- Main issues:
  - Different measurement and dimension
  - Lack of coherence in data
  - No “single version of the truth”



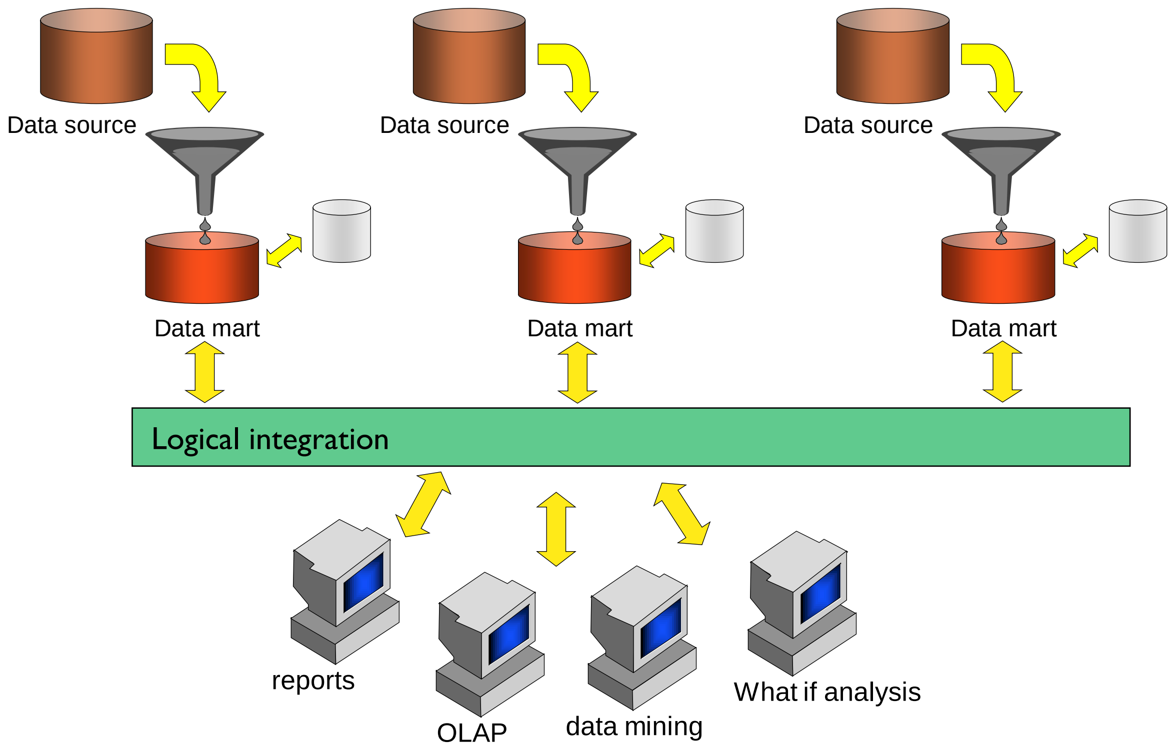
DATA MART BUS

- Data Mart that are logically integrated by common dimensions
- “Enterprise view”
- Proposed by Kimball



FEDERATION

- Ideal in high dynamic contexts (merger and acquisition)
- Critical the integration phase



MULTIDIMENSIONAL MODEL

QUERY

"Usual query: the amount of revenue of last year, region by region and product by product"

"is there any correlation betewn share values and profit in the last 5 years"

"there are two new therapies, what is the one maximize the probabiltiy of safe life?"

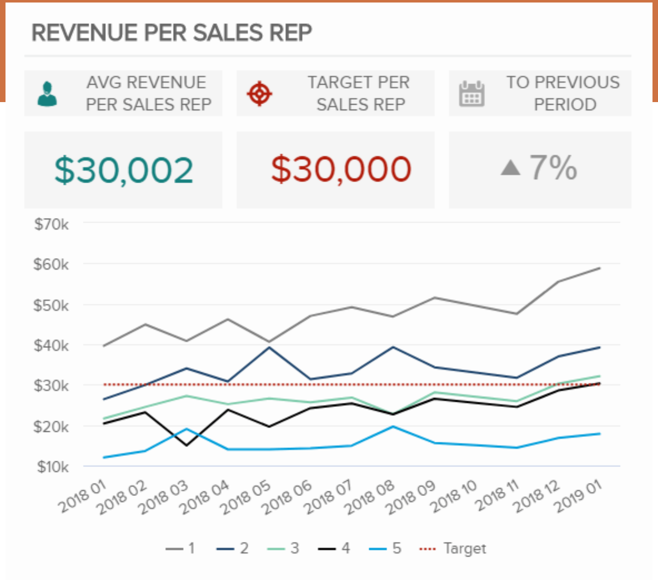
DATA ANALYTICS TECHNIQUES

- Once data has been cleansed, integrated, and transformed, there is the need to figure out how to get the most information out of it
- There are essentially three different approaches, supported by as many categories of tools, to querying a DW by end users:
  - reporting: does not require IT knowledge
  - OLAP: requires the user to think in a multidimensional way and to know the interface of the graphic tool used
  - data mining: requires the user to know the principles that underlie the tools used

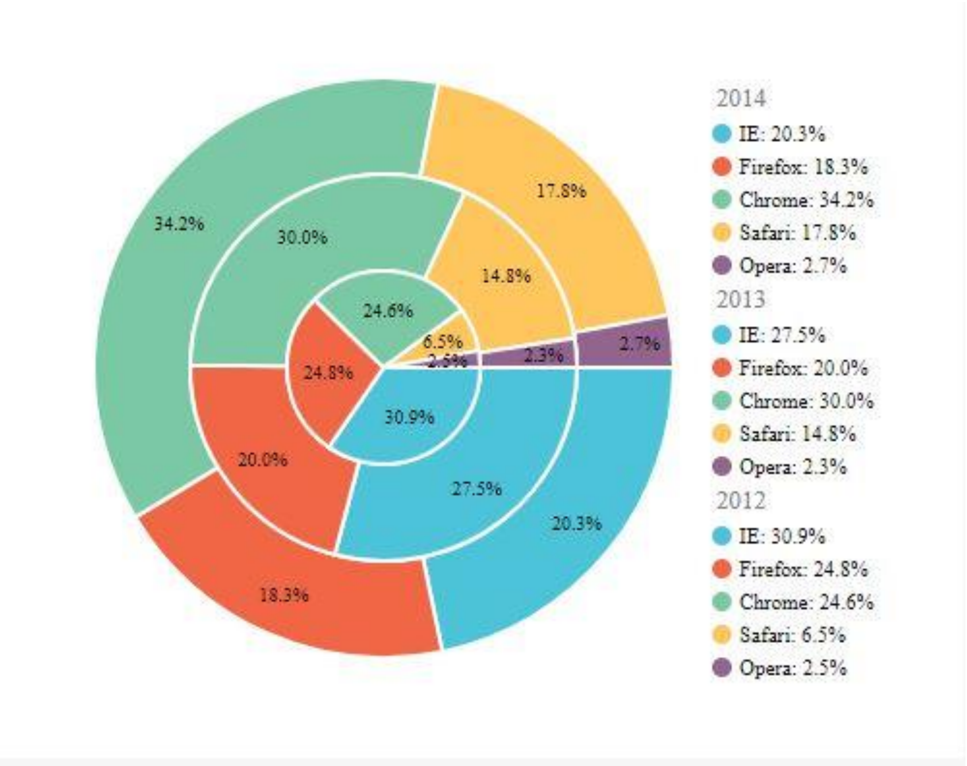
REPORT

User oriented. It provide simple and clear repeated report.

The structure is always the same



Year-Over-Year Growth (YoY) Example: Weekly Retail Unit Sales				
Unit Sales (#)	**2020 Week #31**	**2021 Week #31**	**Change (#)**	**YoY (%)**
Online Sales	5,000	6,000	1,000	**20%**
Retail Sales	2,500	2,000	-500	**-20%**
Total Units Sold	*7,500*	*8,000*	*500*	***7%***



OLAP

- It is the main way of using the information contained in a DW
- It allows users whose analysis needs are not easily identifiable a priori to analyze and explore the data interactively based on the multidimensional model
- While users of reporting tools play an essentially passive role, OLAP users are able to actively build a complex analysis session in which each step taken is a consequence of the results obtained in the previous step
  - Active work sessions
  - in-depth knowledge of the data is required
  - complexity of the queries that can be formulated
  - orientation toward non-IT users

flexible interface, easy to use and effective

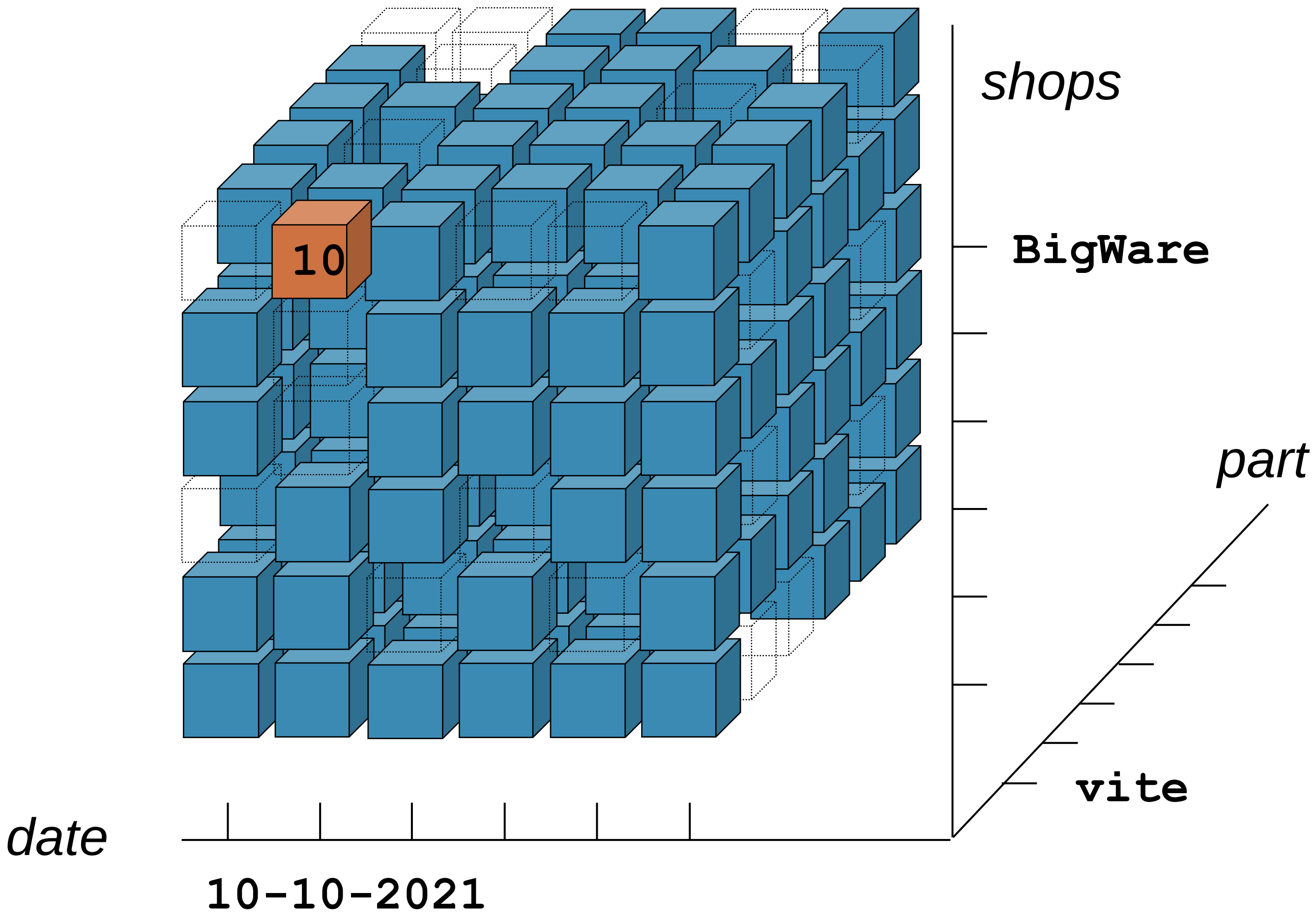
OLAP: SESSION

- An OLAP session consists of a navigation path that reflects the analysis process of one or more facts of interest under different aspects and at different levels of detail. This path takes the form of a sequence of questions often formulated not directly, but by difference with respect to the previous question
- Each step of the analysis session is marked by the application of an OLAP operator which transforms the last query formulated into a new query
- The result of the queries is multidimensional; OLAP tools typically represent data in a tabular fashion highlighting different dimensions using multiple headings, colors, etc.
- OLAP-type navigation is based on a multidimensional data model

THE MULTIDIMENSIONAL MODEL

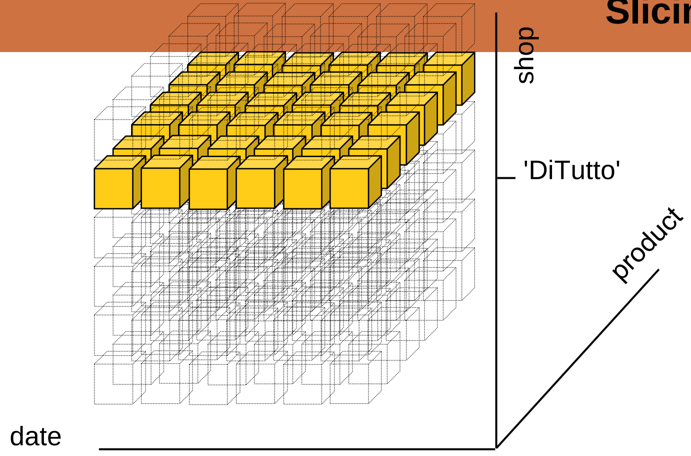
- It is the foundation for representing and querying data in data warehouses
- The facts of interest are represented in cubes where:
  - each cell contains numerical measures which quantify the fact from different points of view;
  - each axis represents a dimension of interest for the analysis;
  - each dimension can be the root of a hierarchy of attributes used to aggregate the data stored in the base cubes.

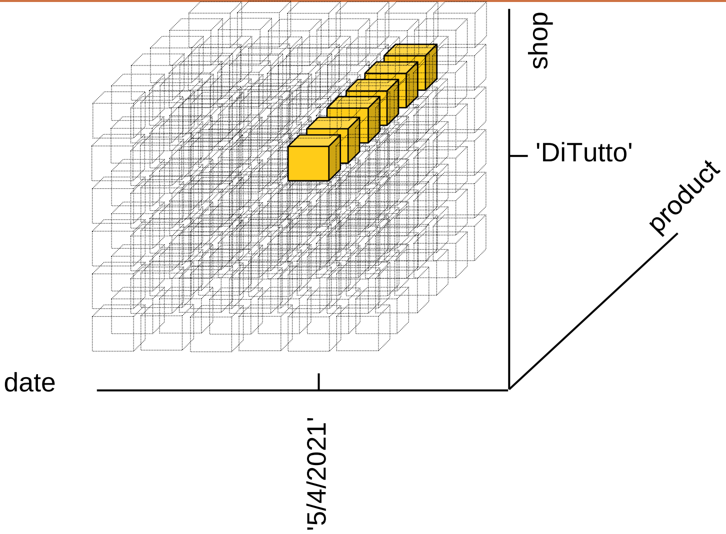
THE SALES CUBE



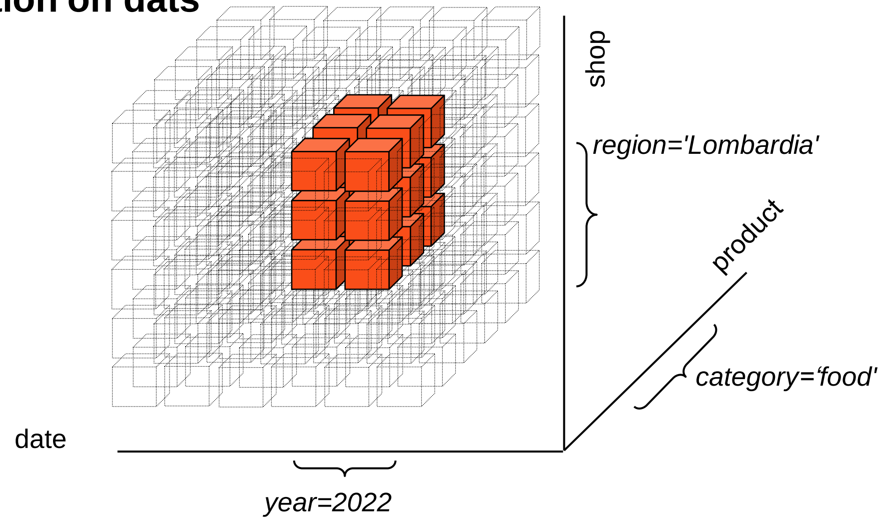
SLICING AND DICING

Slicing = rdata restriction

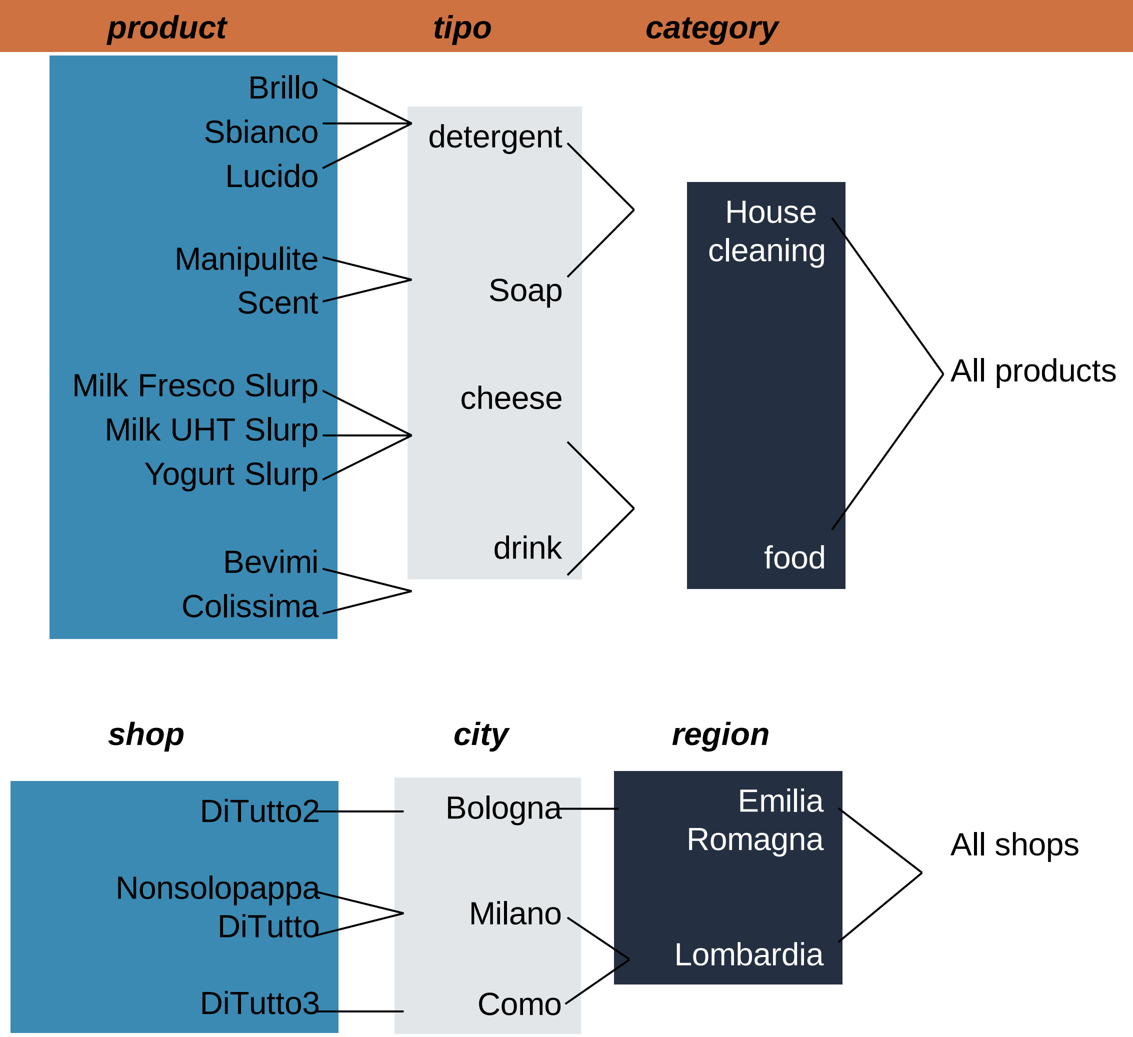




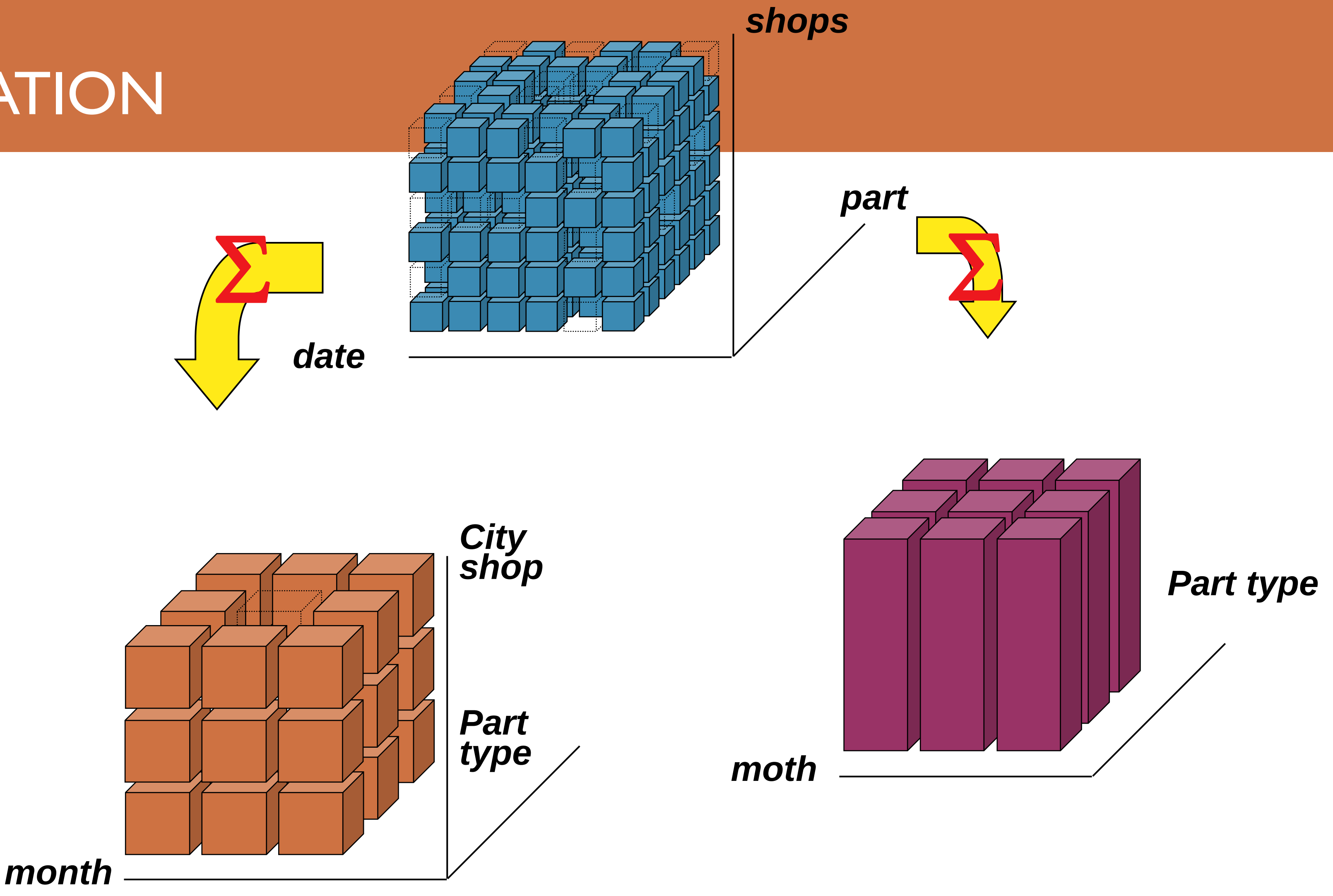
Dicing = condindition on dats



HIERARCHY


product
tipo
category
Brillo
Sbianco
Lucido
Manipulite
Scent
Milk Fresco Slurp
Milk UHT Slurp
Yogurt Slurp
Bevimi
Colissima
detergent
Soap
cheese
drink
House cleaning
food
All products
shop
city
region
DiTutto2
Nonsolopappa
DiTutto
DiTutto3
Bologna
Milano
Como
Emilia Romagna
Lombardia
All shops

AGGREGATION



AGGREGATION

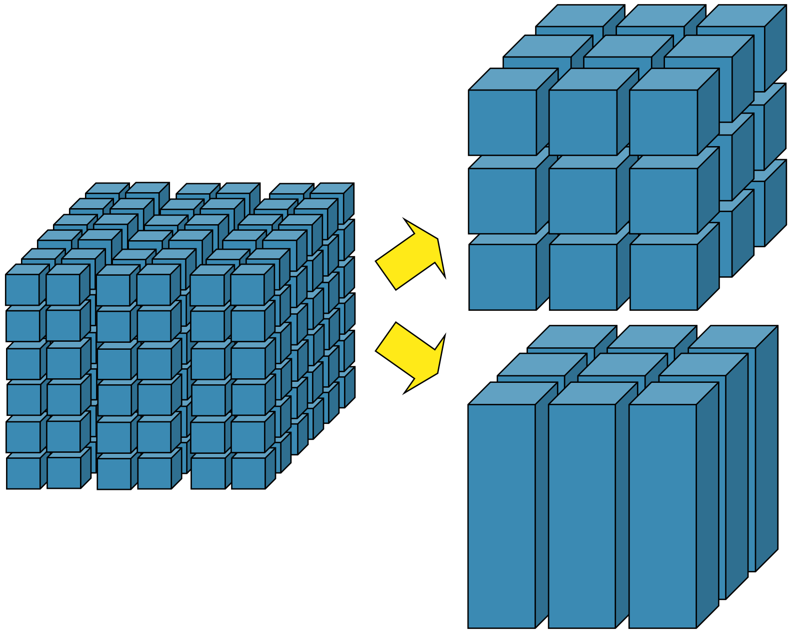
_	DiTutto	DiTutto2	Nonsolopappa
1/1/2020	–	–	–
2/1/2020	10	15	5
3/1/2020	20	–	5
....	....	....	....
1/1/2021	–	–	–
2/1/2021	15	10	20
3/1/2021	20	20	25
....	....	....	....
1/1/2022	–	–	–
2/1/2022	20	8	25
3/1/2022	20	12	20
....	....	....	....

	DiTutto	DiTutto2	Nonsolopappa
Jan 2020	200	180	150
Feb 2020	180	150	120
Mar 2020	220	180	160
....	....	....	....
Jan 2021	350	220	200
Feb 2021	300	200	250
Mar 2021	310	180	300
....	....	....	....
Jan 2022	380	200	220
Feb 2022	310	200	250
Mar 2022	300	160	280
....	....	....	....

	DiTutto	DiTutto2	Nonsolopappa
2020	2400	2000	1600
2021	3200	2300	3000
2022	3400	2200	3200

	DiTutto	DiTutto2	Nonsolopappa
Total	9000	6500	7800

OLAP: TOOLS



roll-up

Roll-Up **:** increase the data aggregation by removing a level of detail from a hierarchy.

OLAP: TOOLS

Month \ Customer Region (Metrics: Dollar Sales)	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Jan 97	$ 620	$ 753	$ 30	$ 660	$ 2.405	$ 1.312	$ 440	$ 1.002	$ 1.002	$ 383	$ 210
Feb 97	$ 258	$ 252	$ 800	$ 975	$ 160	$ 582	$ 744	$ 310	$ 799	$ 118	$ 357
Mar 97	$ 648	$ 244	$ 148	$ 250	$ 1.085	$ 2.961	$ 650	$ 1.240	$ 119	$ 142	$ 96
Apr 97	$ 787	$ 588	$ 447	$ 486	$ 226	$ 506	$ 601	$ 119	$ 550	$ 85	
May 97	$ 1.350	$ 245	$ 936	$ 159	$ 664	$ 626	$ 107	$ 135	$ 200	$ 177	$ 230
Jun 97	$ 842	$ 582	$ 1.281	$ 937	$ 240	$ 774	$ 176	$ 1.139	$ 652	$ 254	$ 745
Jul 97	$ 652	$ 690	$ 486	$ 1.293	$ 605	$ 303	$ 818	$ 103	$ 124	$ 173	$ 66
Aug 97	$ 1.783	$ 304	$ 1.032	$ 170	$ 398	$ 356	$ 432	$ 190	$ 241	$ 407	$ 259
Sep 97	$ 581	$ 778	$ 3.558	$ 587	$ 440	$ 1.652	$ 1.071	$ 315	$ 210	$ 202	
Oct 97	$ 2.291	$ 1.840	$ 600	$ 656	$ 1.300	$ 718	$ 1.210	$ 427	$ 220	$ 520	$ 65
Nov 97	$ 39	$ 1.602	$ 1.082	$ 1.187	$ 842	$ 759	$ 745	$ 232	$ 101	$ 1.037	$ 37
Dec 97	$ 381	$ 1.588	$ 343	$ 118	$ 1.459	$ 635	$ 2.021	$ 259	$ 210	$ 119	$ 189
Jan 98	$ 311	$ 1.174	$ 2.634	$ 3.130	$ 954	$ 2.083	$ 1.351	$ 747	$ 426	$ 447	$ 1.141
Feb 98	$ 2.518	$ 702	$ 1.123	$ 1.336	$ 1.227	$ 3.887	$ 545	$ 268	$ 277	$ 282	
Mar 98	$ 2.459	$ 1.523	$ 1.178	$ 4.708	$ 1.420	$ 3.514	$ 1.948	$ 1.705	$ 276	$ 1.168	$ 63
Apr 98	$ 407	$ 841	$ 524	$ 712	$ 133	$ 2.486	$ 49	$ 390	$ 1.298	$ 221	$ 46
May 98	$ 667	$ 1.721	$ 440	$ 148	$ 80	$ 1.310	$ 303	$ 104	$ 657	$ 65	
Jun 98	$ 699	$ 1.096	$ 898	$ 353	$ 902	$ 839		$ 230	$ 155	$ 105	$ 75
Jul 98	$ 586	$ 1.897	$ 412	$ 226	$ 406	$ 361	$ 1.628	$ 267	$ 1.011	$ 41	$ 184
Aug 98	$ 894	$ 326	$ 792	$ 1.832	$ 1.199	$ 295	$ 1.816	$ 277	$ 102	$ 118	$ 115
Sep 98	$ 338	$ 3.179	$ 505	$ 427	$ 99	$ 2.976	$ 885	$ 135	$ 85	$ 1.110	$ 510
Oct 98	$ 544	$ 413	$ 1.467	$ 209	$ 679	$ 706	$ 556	$ 480	$ 485	$ 99	$ 160
Nov 98	$ 671	$ 459	$ 1.471	$ 2.066	$ 701	$ 716	$ 986	$ 1.127	$ 154	$ 440	$ 361
Dec 98	$ 836	$ 2.096	$ 1.726	$ 3.642	$ 395	$ 1.740	$ 1.943	$ 1.143	$ 366	$ 307	$ 118

roll-up

Quarter \ Customer Region (Metrics: Dollar Sales)	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Q1 1997	$ 1.526	$ 1.249	$ 978	$ 1.885	$ 3.650	$ 4.855	$ 1.834	$ 2.552	$ 1.920	$ 643	$ 663
Q2 1997	$ 2.979	$ 1.415	$ 2.664	$ 1.582	$ 1.130	$ 1.906	$ 884	$ 1.393	$ 1.402	$ 516	$ 975
Q3 1997	$ 3.016	$ 1.772	$ 5.076	$ 2.050	$ 1.443	$ 2.311	$ 2.321	$ 608	$ 575	$ 782	$ 325
Q4 1997	$ 2.711	$ 5.030	$ 2.025	$ 1.961	$ 3.601	$ 2.112	$ 3.976	$ 918	$ 531	$ 1.676	$ 291
Q1 1998	$ 5.288	$ 3.399	$ 4.935	$ 9.174	$ 3.601	$ 9.484	$ 3.844	$ 2.720	$ 979	$ 1.897	$ 1.204
Q2 1998	$ 1.773	$ 3.658	$ 1.862	$ 1.213	$ 1.115	$ 4.635	$ 352	$ 724	$ 2.110	$ 391	$ 121
Q3 1998	$ 1.818	$ 5.402	$ 1.709	$ 2.485	$ 1.704	$ 3.632	$ 4.329	$ 679	$ 1.198	$ 1.269	$ 809
Q4 1998	$ 2.051	$ 2.968	$ 4.664	$ 5.917	$ 1.775	$ 3.162	$ 3.485	$ 2.750	$ 1.005	$ 846	$ 639

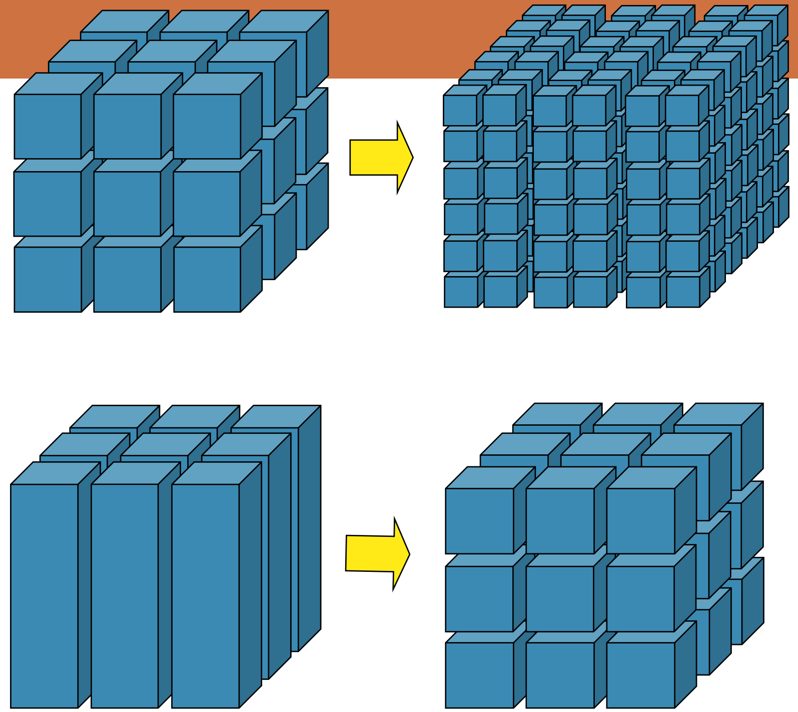
OLAP: TOOLS

Category	Year	Metrics / Customer Region	Dollar Sales North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germa
Electronics	1997		$ 138	$ 1.774	$ 384	$ 138	$ 2.346	$ 2.554	$ 2.184	$ 566	$ 199	$
	1998		$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462	$ 7
Food	1997		$ 759	$ 682	$ 729	$ 262	$ 588	$ 469	$ 807	$ 156	$ 615	$ 1
	1998		$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175	$ 1
Gifts	1997		$ 2.532	$ 1.355	$ 1.854	$ 1.413	$ 2.535	$ 2.132	$ 1.904	$ 908	$ 375	$ 1.0
	1998		$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717	$ 6
Health & Beauty	1997		$ 624	$ 640	$ 1.317	$ 647	$ 588	$ 754	$ 654	$ 143	$ 292	$ 3
	1998		$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72	
Household	1997		$ 5.354	$ 4.112	$ 5.410	$ 4.446	$ 3.058	$ 3.974	$ 2.654	$ 3.545	$ 2.875	$ 1.9
	1998		$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649	$ 2.7
Kid's Korner	1997		$ 201	$ 398	$ 485	$ 186	$ 409	$ 323	$ 396	$ 105	$ 34	$
	1998		$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19	$
Travel	1997		$ 624	$ 505	$ 564	$ 386	$ 300	$ 978	$ 416	$ 48	$ 38	
	1998		$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198	$

roll-up

Category	Year	Metrics	Dollar Sales
Electronics	1997		$ 10.616
	1998		$ 29.299
Food	1997		$ 5.300
	1998		$ 5.638
Gifts	1997		$ 16.315
	1998		$ 20.047
Health & Beauty	1997		$ 6.042
	1998		$ 5.665
Household	1997		$ 38.383
	1998		$ 50.391
Kid's Korner	1997		$ 2.559
	1998		$ 2.943
Travel	1997		$ 4.497
	1998		$ 4.792

OLAP: TOOLS



drill-down

Drill-Down **:** decrease data aggregation by introducing an additional level of detail into a hierarchy

OLAP:TOOLS

Metrics	Dollar Sales										
Customer Region	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Quarter											
Q1 1997	$ 1.526	$ 1.249	$ 978	$ 1.885	$ 3.650	$ 4.855	$ 1.834	$ 2.552	$ 1.920	$ 643	$ 663
Q2 1997	$ 2.979	$ 1.415	$ 2.664	$ 1.582	$ 1.130	$ 1.906	$ 884	$ 1.393	$ 1.402	$ 516	$ 975
Q3 1997	$ 3.016	$ 1.772	$ 5.076	$ 2.050	$ 1.443	$ 2.311	$ 2.321	$ 608	$ 575	$ 782	$ 325
Q4 1997	$ 2.711	$ 5.030	$ 2.025	$ 1.961	$ 3.601	$ 2.112	$ 3.976	$ 918	$ 531	$ 1.676	$ 291
Q1 1998	$ 5.288	$ 3.399	$ 4.935	$ 9.174	$ 3.601	$ 9.484	$ 3.844	$ 2.720	$ 979	$ 1.897	$ 1.204
Q2 1998	$ 1.773	$ 3.658	$ 1.862	$ 1.213	$ 1.115	$ 4.635	$ 352	$ 724	$ 2.110	$ 391	$ 121
Q3 1998	$ 1.818	$ 5.402	$ 1.709	$ 2.485	$ 1.704	$ 3.632	$ 4.329	$ 679	$ 1.198	$ 1.269	$ 809
Q4 1998	$ 2.051	$ 2.968	$ 4.664	$ 5.917	$ 1.775	$ 3.162	$ 3.485	$ 2.750	$ 1.005	$ 846	$ 639

drill-down

Metrics	Dollar Sales													
Customer City	Arlin	San Pedro	Springfield	Chappel Hill	Scranburg	Pebble Beach	Martinsville	Maddon	Peoria	Pecos	Lake Barkley	Alcameda	Fingers Lake	S
Quarter														
Q1 1997	$ 675										$ 39			
Q2 1997				$ 203					$ 53				$ 135	
Q3 1997				$ 276								$ 252	$ 63	
Q4 1997	$ 215	$ 124			$ 113	$ 45	$ 192	$ 348				$ 79	$ 98	
Q1 1998			$ 140	$ 174			$ 85				$ 237	$ 30	$ 119	
Q2 1998								$ 12	$ 17					
Q3 1998	$ 734					$ 25	$ 1.535							
Q4 1998						$ 219	$ 119	$ 142		$ 85	$ 1.533			

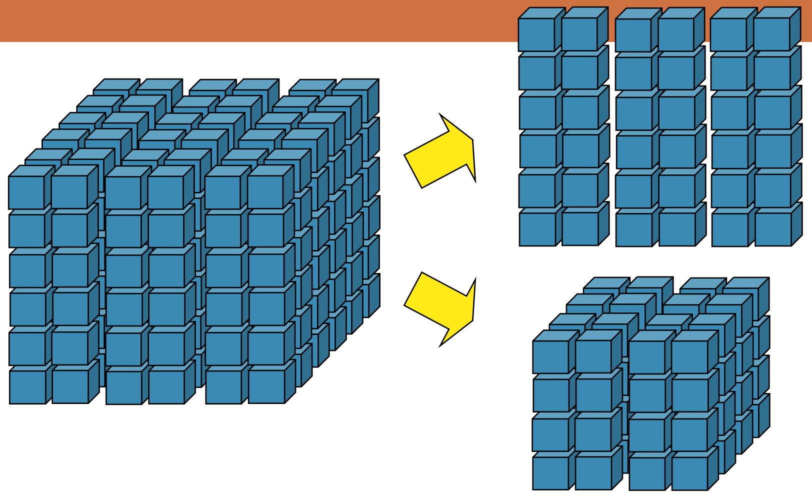
OLAP: TOOLS

Metrics	Dollar Sales	
Year	1997	1998
Category		
Electronics	$ 10.616	$ 29.299
Food	$ 5.300	$ 5.638
Gifts	$ 16.315	$ 20.047
Health & Beauty	$ 6.042	$ 5.665
Household	$ 38.383	$ 50.391
Kid's Korner	$ 2.559	$ 2.943
Travel	$ 4.497	$ 4.792

drill-down

Metrics	Dollar Sales											
Customer Region	North-East		Mid-Atlantic		South-East		Central		South		North-West	
Year	1997	1998	1997	1998	1997	1998	1997	1998	1997	1998	1997	1998
Category												
Electronics	$ 138	$ 1.184	$ 1.774	$ 4.529	$ 384	$ 1.892	$ 138	$ 7.232	$ 2.346	$ 651	$ 2.554	$ 9.488
Food	$ 759	$ 538	$ 682	$ 925	$ 729	$ 959	$ 262	$ 677	$ 588	$ 213	$ 469	$ 1.503
Gifts	$ 2.532	$ 1.955	$ 1.355	$ 2.785	$ 1.854	$ 2.800	$ 1.413	$ 2.695	$ 2.535	$ 1.813	$ 2.132	$ 2.844
Health & Beauty	$ 624	$ 611	$ 640	$ 887	$ 1.317	$ 566	$ 647	$ 382	$ 588	$ 499	$ 754	$ 1.162
Household	$ 5.354	$ 5.787	$ 4.112	$ 5.320	$ 5.410	$ 5.416	$ 4.446	$ 6.812	$ 3.058	$ 4.334	$ 3.974	$ 5.008
Kid's Korner	$ 201	$ 247	$ 398	$ 422	$ 485	$ 441	$ 186	$ 380	$ 409	$ 221	$ 323	$ 592
Travel	$ 624	$ 608	$ 505	$ 559	$ 564	$ 1.096	$ 386	$ 611	$ 300	$ 464	$ 978	$ 316

OLAP: TOOLS



slice-and-dice

Slice-and-Dice **:** (tagliare a fette e a cubetti)

✓ **Slice :** riduce la dimensionalità del cubo fissando un valore per una delle dimensioni;

✓ **Dice :** (selezione) riduce l’insieme dei dati attraverso la formulazione di un criterio di selezione.

OLAP: TOOLS

Category	Year	Metrics / Customer Region	Dollar Sales: North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germa
Electronics	1997		$ 138	$ 1.774	$ 384	$ 138	$ 2.346	$ 2.554	$ 2.184	$ 566	$ 199	$
	1998		$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462	$ 7
Food	1997		$ 759	$ 682	$ 729	$ 262	$ 588	$ 469	$ 807	$ 156	$ 615	$ 1
	1998		$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175	$ 1
Gifts	1997		$ 2.532	$ 1.355	$ 1.854	$ 1.413	$ 2.535	$ 2.132	$ 1.904	$ 908	$ 375	$ 1.0
	1998		$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717	$ 6
Health & Beauty	1997		$ 624	$ 640	$ 1.317	$ 647	$ 588	$ 754	$ 654	$ 143	$ 292	$ 3
	1998		$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72	
Household	1997		$ 5.354	$ 4.112	$ 5.410	$ 4.446	$ 3.058	$ 3.974	$ 2.654	$ 3.545	$ 2.875	$ 1.9
	1998		$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649	$ 2.7
Kid's Korner	1997		$ 201	$ 398	$ 485	$ 186	$ 409	$ 323	$ 396	$ 105	$ 34	$
	1998		$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19	$
Travel	1997		$ 624	$ 505	$ 564	$ 386	$ 300	$ 978	$ 416	$ 48	$ 38	
	1998		$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198	$

Filter Details:
Year = 1998

Category	Metrics / Customer Region	Dollar Sales: North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Ca
Electronics		$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462	$ 702	
Food		$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175	$ 100	$
Gifts		$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717	$ 686	$
Health & Beauty		$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72		$
Household		$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649	$ 2.791	$
Kid's Korner		$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19	$ 69	
Travel		$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198	$ 55	

Slicing Year=1998

OLAP: TOOLS

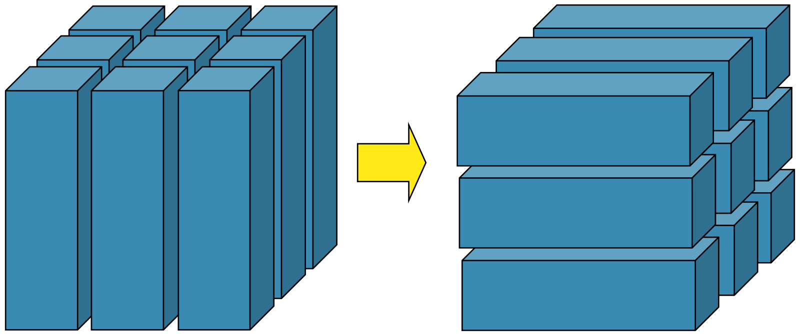
Metrics	Dollar Sales											
Customer City	Afton	Akron	Albon	Alcameda	Alka	Allagash	Alta	Altoola	Amestra	Amsterdam	Andersonville	Annap
Subcategory												
Audio						$ 85						
Automotive								$ 30				
Chocolate	$ 42	$ 42		$ 50		$ 20		$ 22	$ 44			$
Christmas	$ 30					$ 25	$ 30	$ 15				
Classic Toys						$ 7	$ 26				$ 38	
Coffee			$ 9									
Comfort				$ 59		$ 59						
Furniture							$ 485					
Gadgets							$ 199	$ 79	$ 79			
Games & Puzzles							$ 17		$ 45		$ 45	
Gift Baskets			$ 55	$ 43								$
Golf	$ 25							$ 25	$ 14		$ 25	
Hearth									$ 15			
Jewelry	$ 75			$ 189		$ 24	$ 77	$ 189	$ 24			
Kitchen						$ 55	$ 21		$ 76			$
Lawn & Garden	$ 75		$ 100		$ 15	$ 63	$ 100		$ 180	$ 67	$ 40	$
Learning	$ 16							$ 37				
Meat & Cheese		$ 40		$ 20			$ 20				$ 25	
Miscellaneous		$ 200	$ 1.320		$ 200	$ 139			$ 993			
Natural Remedies	$ 13								$ 13			
Pets	$ 215		$ 26			$ 30	$ 68	$ 115	$ 25		$ 34	$
Plants & Flowers	$ 65	$ 65	$ 65				$ 50	$ 60				$
Safety & Security								$ 30	$ 22	$ 22		
Skin Care												
Sleeping			$ 18									
Toys & Accessories							$ 29	$ 185	$ 744			$

Dicing over complex predicate

Filter Details:
Category = Electronics
AND
Dollar Sales > 80
AND
Customer Region = North-West
AND
Year = 1997

Metrics	Dollar Sales					
Customer City	Alta	Armstrong	Avery Heights	Lane	Mt. Everest	San Fransisco
Subcategory						
Audio		$ 98		$ 123	$ 85	
Comfort			$ 118		$ 1.495	
Gadgets	$ 199					$ 199

OLAP: TOOLS



pivoting

Pivoting **:** change the way of presentation with the aim of analyzing the same information from different points of view

OLAP: TOOLS

Category	Year	Metrics: Dollar Sales
Electronics	1997	$ 10.616
	1998	$ 29.299
Food	1997	$ 5.300
	1998	$ 5.638
Gifts	1997	$ 16.315
	1998	$ 20.047
Health & Beauty	1997	$ 6.042
	1998	$ 5.665
Household	1997	$ 38.383
	1998	$ 50.391
Kid's Korner	1997	$ 2.559
	1998	$ 2.943
Travel	1997	$ 4.497
	1998	$ 4.792

pivoting

Metrics	Dollar Sales	
Year	1997	1998
Category		
Electronics	$ 10.616	$ 29.299
Food	$ 5.300	$ 5.638
Gifts	$ 16.315	$ 20.047
Health & Beauty	$ 6.042	$ 5.665
Household	$ 38.383	$ 50.391
Kid's Korner	$ 2.559	$ 2.943
Travel	$ 4.497	$ 4.792

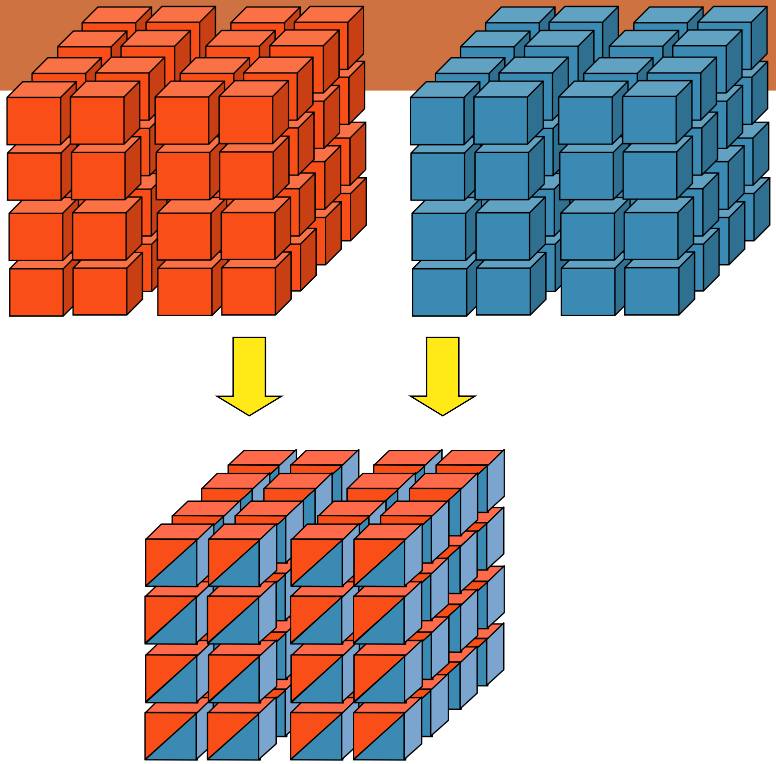
OLAP: TOOLS

Category	Year	Dollar Sales North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germa
Electronics	1997	$ 138	$ 1.774	$ 384	$ 138	$ 2.346	$ 2.554	$ 2.184	$ 566	$ 199	$
	1998	$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462	$ 7
Food	1997	$ 759	$ 682	$ 729	$ 262	$ 588	$ 469	$ 807	$ 156	$ 615	$ 1
	1998	$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175	$ 1
Gifts	1997	$ 2.532	$ 1.355	$ 1.854	$ 1.413	$ 2.535	$ 2.132	$ 1.904	$ 908	$ 375	$ 1.0
	1998	$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717	$ 6
Health & Beauty	1997	$ 624	$ 640	$ 1.317	$ 647	$ 588	$ 754	$ 654	$ 143	$ 292	$ 3
	1998	$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72	
Household	1997	$ 5.354	$ 4.112	$ 5.410	$ 4.446	$ 3.058	$ 3.974	$ 2.654	$ 3.545	$ 2.875	$ 1.9
	1998	$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649	$ 2.7
Kid's Korner	1997	$ 201	$ 398	$ 485	$ 186	$ 409	$ 323	$ 396	$ 105	$ 34	$
	1998	$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19	$
Travel	1997	$ 624	$ 505	$ 564	$ 386	$ 300	$ 978	$ 416	$ 48	$ 38	
	1998	$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198	$

pivoting

Metrics / Customer Region / Year	Dollar Sales North-East 1997	North-East 1998	Mid-Atlantic 1997	Mid-Atlantic 1998	South-East 1997	South-East 1998	Central 1997	Central 1998	South 1997	South 1998	North-West 1997	North-West 1998
Category												
Electronics	$ 138	$ 1.184	$ 1.774	$ 4.529	$ 384	$ 1.892	$ 138	$ 7.232	$ 2.346	$ 651	$ 2.554	$ 9.488
Food	$ 759	$ 538	$ 682	$ 925	$ 729	$ 959	$ 262	$ 677	$ 588	$ 213	$ 469	$ 1.503
Gifts	$ 2.532	$ 1.955	$ 1.355	$ 2.785	$ 1.854	$ 2.800	$ 1.413	$ 2.695	$ 2.535	$ 1.813	$ 2.132	$ 2.844
Health & Beauty	$ 624	$ 611	$ 640	$ 887	$ 1.317	$ 566	$ 647	$ 382	$ 588	$ 499	$ 754	$ 1.162
Household	$ 5.354	$ 5.787	$ 4.112	$ 5.320	$ 5.410	$ 5.416	$ 4.446	$ 6.812	$ 3.058	$ 4.334	$ 3.974	$ 5.008
Kid's Korner	$ 201	$ 247	$ 398	$ 422	$ 485	$ 441	$ 186	$ 380	$ 409	$ 221	$ 323	$ 592
Travel	$ 624	$ 608	$ 505	$ 559	$ 564	$ 1.096	$ 386	$ 611	$ 300	$ 464	$ 978	$ 316

OLAP: TOOLS



drill-across

Drill-Across **:** it creates a join between two cubes

OLAP: TOOLS

	Metrics	Dollar Sales							
	Quarter	Q1 1997	Q2 1997	Q3 1997	Q4 1997	Q1 1998	Q2 1998	Q3 1998	Q4 1998
Category									
Electronics		$ 4.383	$ 817	$ 827	$ 4.589	$ 13.770	$ 2.977	$ 4.226	$ 8.326
Food		$ 1.546	$ 1.310	$ 1.268	$ 1.176	$ 2.676	$ 1.120	$ 953	$ 889
Gifts		$ 3.398	$ 3.893	$ 4.682	$ 4.342	$ 7.879	$ 4.145	$ 4.378	$ 3.645
Health & Beauty		$ 1.826	$ 878	$ 1.904	$ 1.434	$ 2.156	$ 898	$ 1.207	$ 1.404
Household		$ 9.314	$ 8.124	$ 9.331	$ 11.614	$ 17.453	$ 7.604	$ 12.898	$ 12.436
Kid's Korner		$ 685	$ 531	$ 811	$ 532	$ 1.084	$ 491	$ 532	$ 836
Travel		$ 603	$ 1.293	$ 1.456	$ 1.145	$ 1.507	$ 719	$ 840	$ 1.726

drill-across

Quarter	Q1 1997		Q2 1997		Q3 1997		Q4 1997		Q1 1998		Q2 1998	
Metrics	Discount	Dollar Sales	Discount	Dollar Sales	Discount	Dollar Sales	Discount	Dollar Sales	Discount	Dollar Sales	Discount	Dollar Sales
Category												
Electronics	$ 0	$ 4.383	$ 0	$ 817	$ 0	$ 827	$ 300	$ 4.589	$ 15	$ 13.770	$ 0	$ 2.97
Food	$ 25	$ 1.546	$ 0	$ 1.310	$ 0	$ 1.268	$ 38	$ 1.176	$ 0	$ 2.676	$ 0	$ 1.12
Gifts	$ 31	$ 3.398	$ 0	$ 3.893	$ 5	$ 4.682	$ 0	$ 4.342	$ 15	$ 7.879	$ 0	$ 4.14
Health & Beauty	$ 0	$ 1.826	$ 0	$ 878	$ 0	$ 1.904	$ 0	$ 1.434	$ 229	$ 2.156	$ 0	$ 89
Household	$ 0	$ 9.314	$ 228	$ 8.124	$ 175	$ 9.331	$ 35	$ 11.614	$ 5	$ 17.453	$ 211	$ 7.60
Kid's Korner	$ 0	$ 685	$ 0	$ 531	$ 32	$ 811	$ 40	$ 532	$ 0	$ 1.084	$ 0	$ 49
Travel	$ 0	$ 603	$ 0	$ 1.293	$ 200	$ 1.456	$ 0	$ 1.145	$ 0	$ 1.507	$ 0	$ 71

Drill-across between Dollar Sales cube and Discount cube.

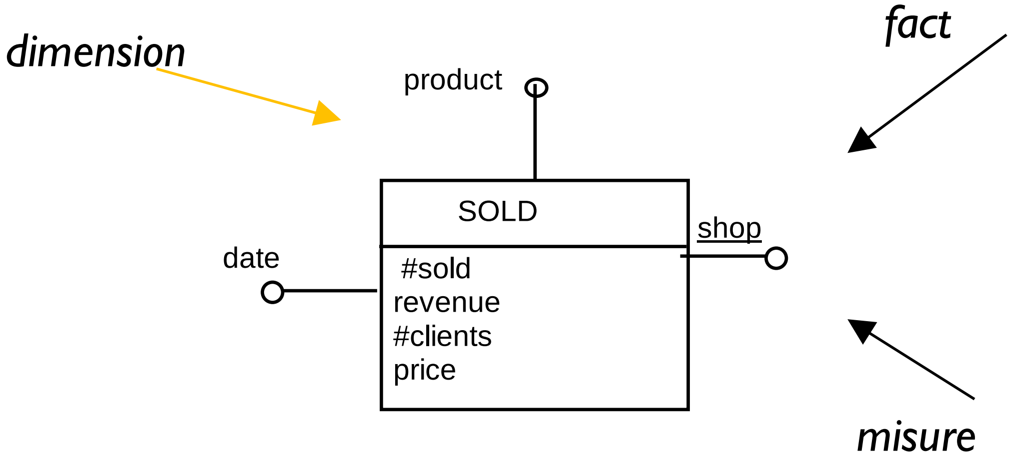
DIMENSIONAL FACT MODEL (DFM)

IL DIMENSIONAL FACT MODEL

- The DFM is a graphical conceptual model for data marts, designed to:
  - effectively support the conceptual design;
  - create an environment on which to intuitively formulate user queries;
  - allow dialogue between the designer and the end user to refine the specifications of the requirements;
  - create a stable platform from which to start the logic project (regardless of the target logic model);
  - return expressive and unambiguous a posteriori documentation.
- The conceptual representation generated by the DFM consists of a set of fact schemas. The basic elements modeled by fact schemas are facts, measures, dimensions, and hierarchies

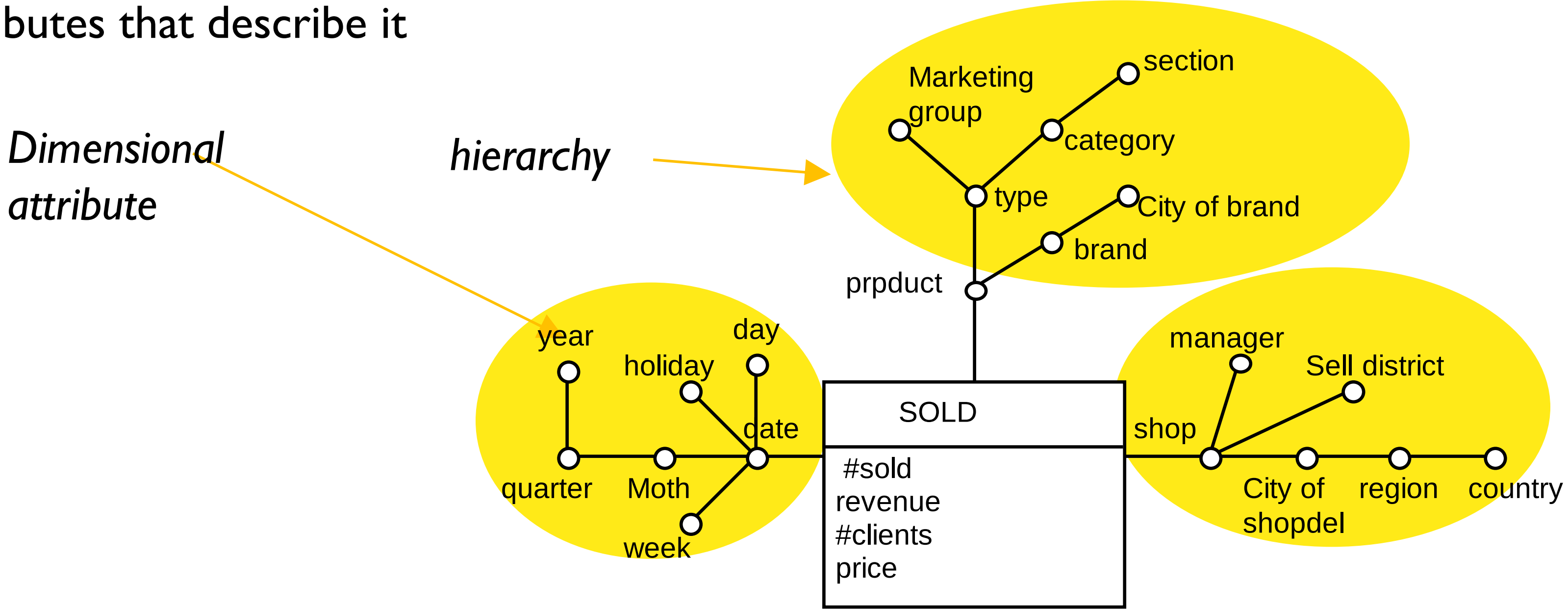
IL DFM: BASIC CONSTRUCTS

- A fact is a concept of interest for decision making; typically models a set of events that occur in the enterprise (for example: sales, shipments, purchases, ...). It is essential that a fact has dynamic aspects, that is, it evolves over time
- A measure is a numerical property of a fact and describes a quantitative aspect of interest for the analysis (for example, each sale is measured by its receipts)
- A dimension is a finite-domain property of a fact and describes an analysis coordinate (typical dimensions for the sales fact are product, store, date)



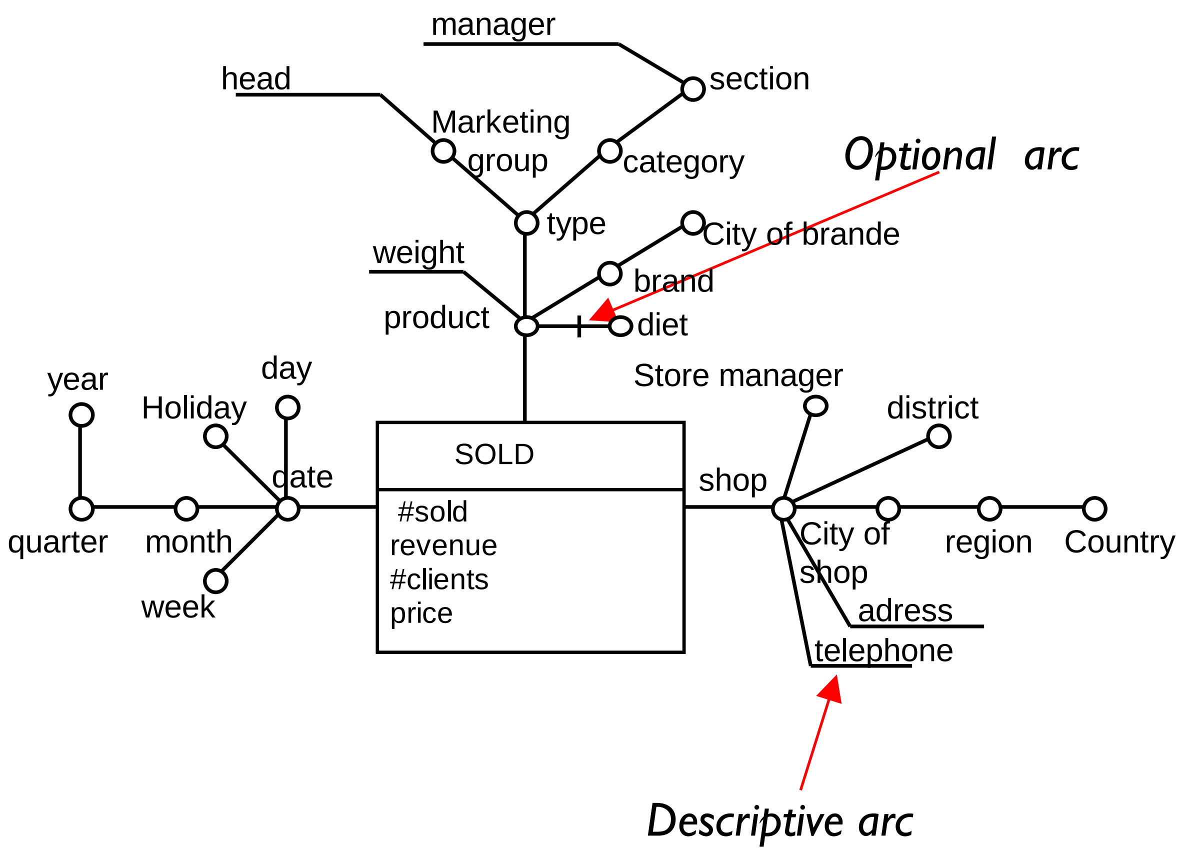
DFM: BASIC CONSTRUCT

- The general term dimensional attributes means the dimensions and any other attributes, always with discrete values, which describe them (for example, a product is described by its type, by the category to which it belongs, by its brand, by the department in which it is sold)
- A hierarchy is a directed tree whose nodes are dimensional attributes and whose edges model many-to-one associations between pairs of dimensional attributes. It contains a dimension, located at the root of the tree, and all the dimensional attributes that describe it



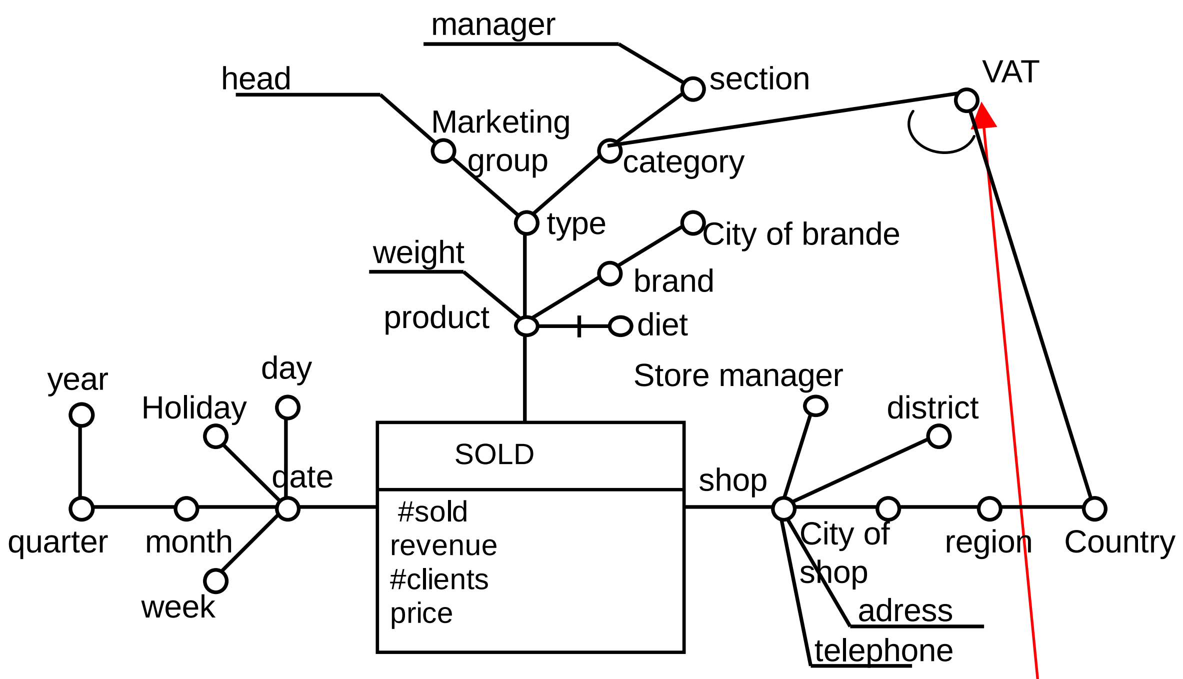
DFM: ADVANCED

- A descriptive attribute contains additional information about a dimensional attribute of a hierarchy, to which it is connected by a one-to-one association. It is not used for aggregation because it has continuous values and/or because it comes from a one-to-one association
- Some arcs of the fact schema may be optional



DFM: ADVANCED MODELLING

- A cross-dimensional attribute is an attribute, dimensional or descriptive, whose value is determined by the combination of two or more dimensional attributes, possibly belonging to distinct hierarchies

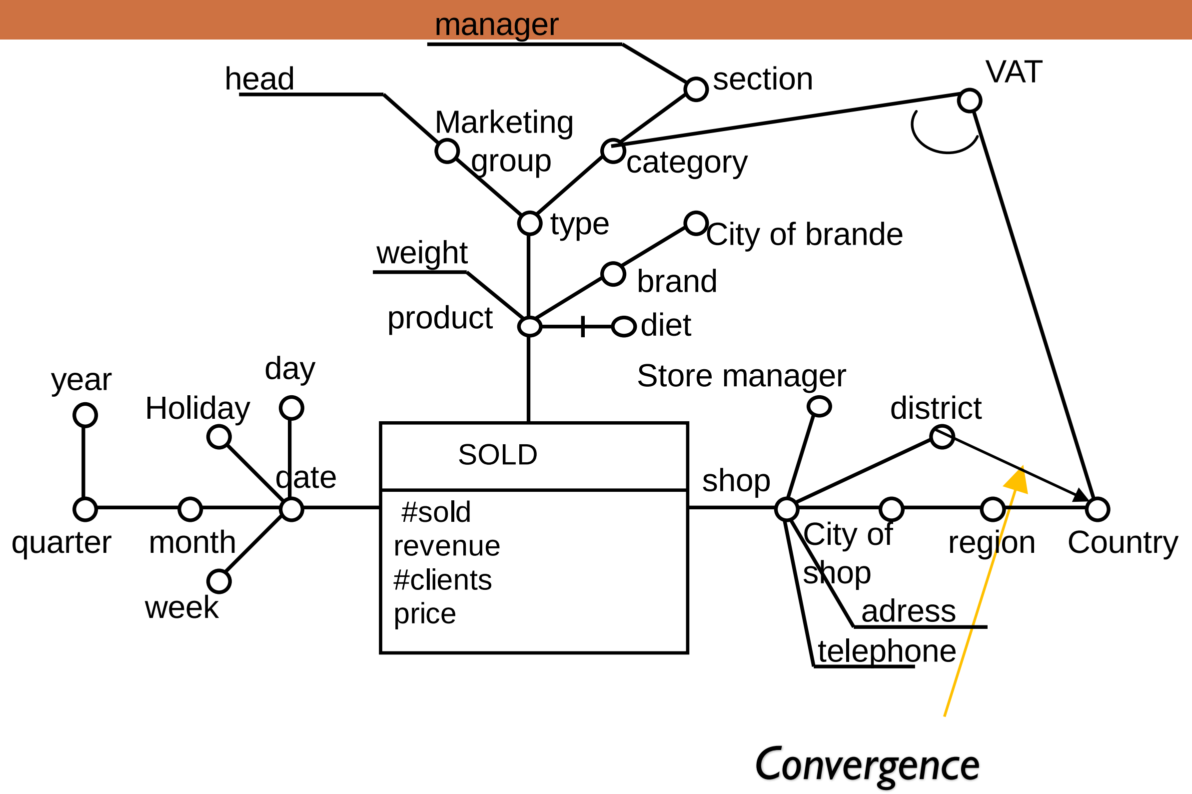


cross-dimensional attribute

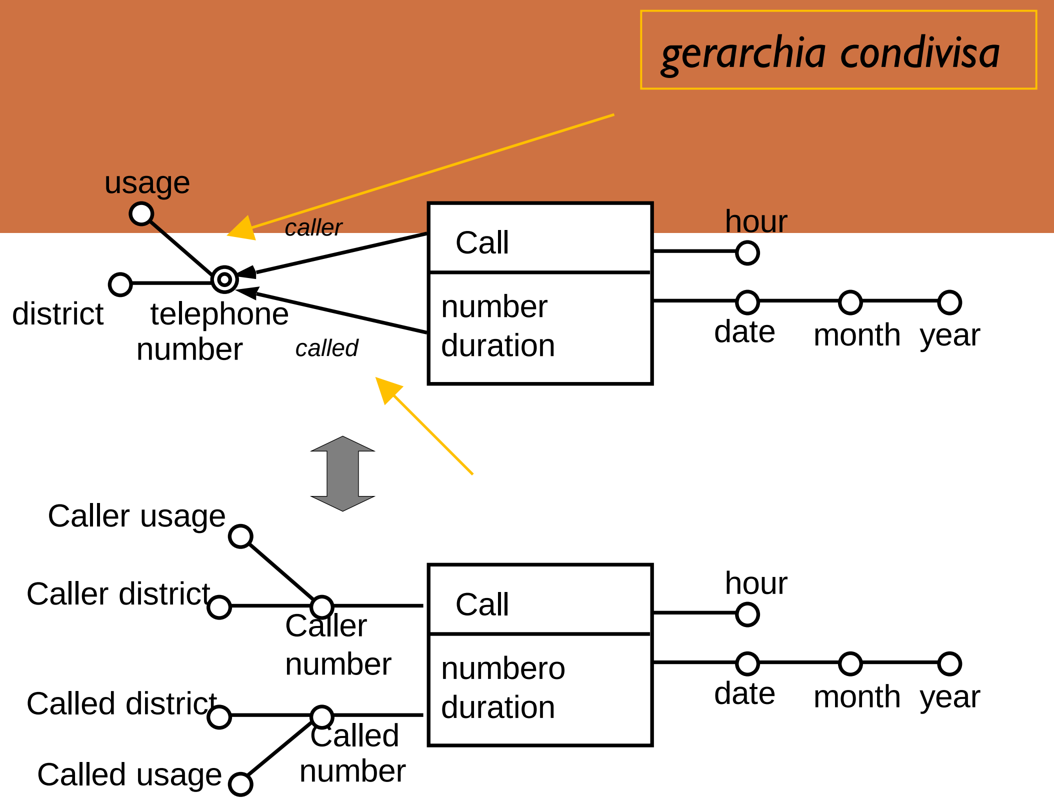
- Another example: public holiday

IL DFM: ADVANCED

- Two dimensional attributes can be connected by two or more distinct directed paths, provided that each of them still represents a functional dependency (convergence)



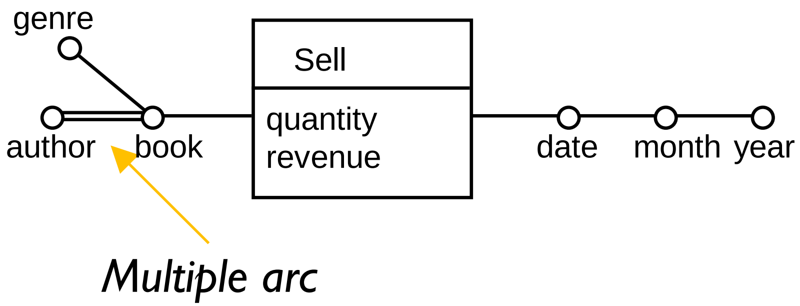
IL DFM: ADVANCED



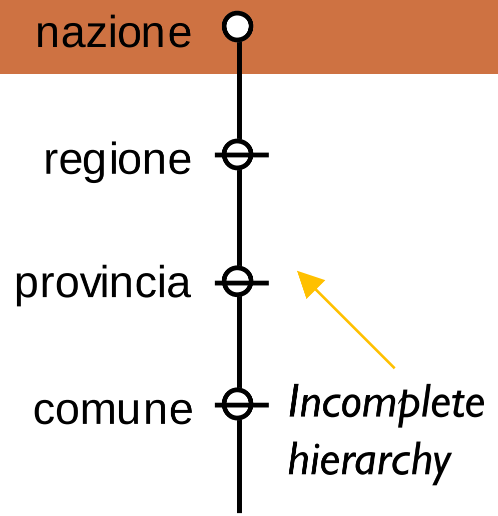
- Shared hierarchy is an abbreviation used to denote that a portion of the hierarchy is replicated multiple times in the schema

IL DFM: COSTRUTTI AVANZATI

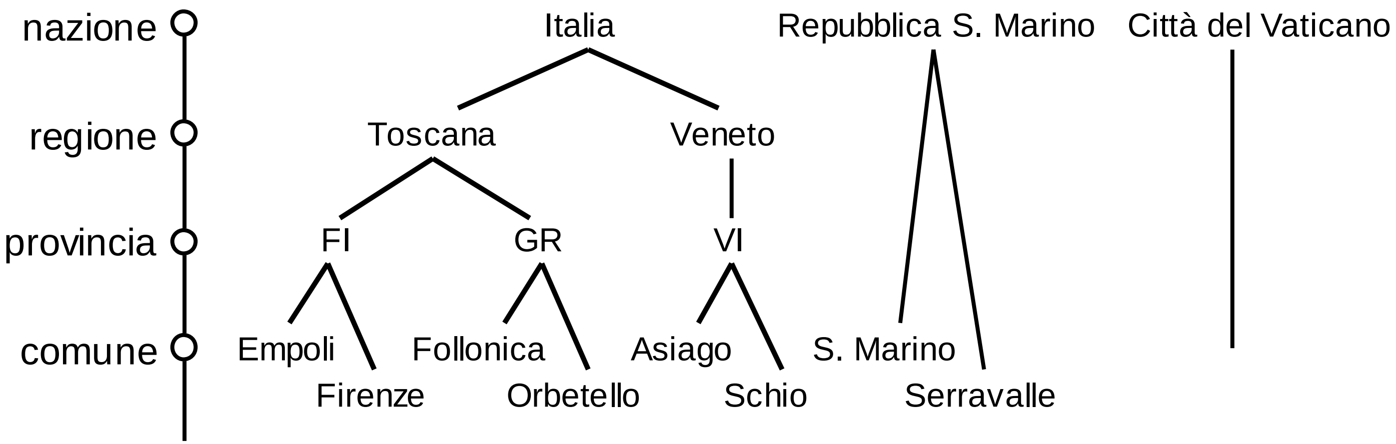
- A multiple arc models a many-to-many association between two dimensional attributes



DFM: ADVANCED MODELL



- An incomplete hierarchy is a hierarchy in which, for some instances, one or more levels of aggregation are absent (because they are not known or not defined).

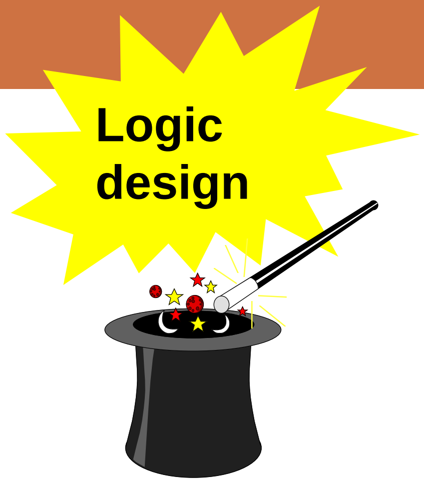


FROM DFM TO STAR SCHEMA

LOGIC DESIGN

INPUT

Conceptual scheme
Workload
Data volume
System constraints



OUTPUT

Logic schema

- The set of steps which, starting from the conceptual scheme, allow to determine the logical scheme of the data mart

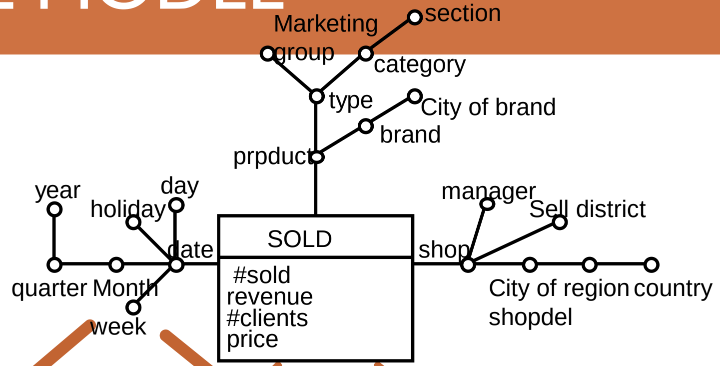
- It is based on different and often inconsistent principles with those used in operational systems
  - Data redundancy
  - Denormalization of relationships

LOGIC DESIGN

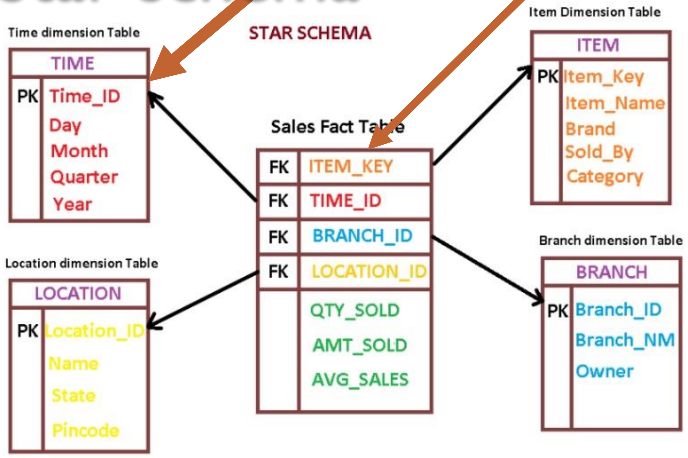
- The main operations to be performed during logical design are:

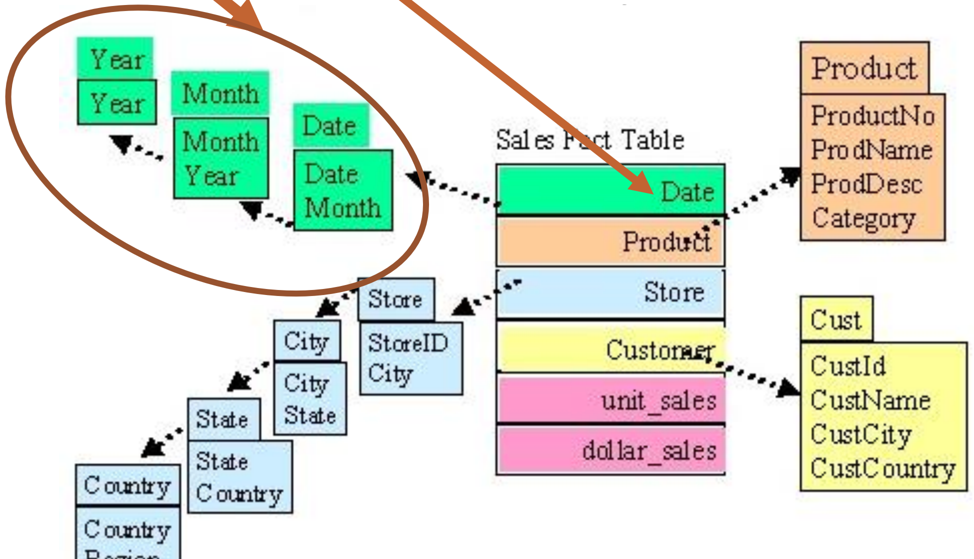
1. Choice of logical schema to use (e.g. star/snowflake schema)
2. Translation of conceptual schemes
3. Choice of dynamic hierarchies
4. Choice of views to materialize
5. Applying other forms of optimization (e.g. vertical/horizontal fragmentation)

FROM DFM TO LOGICAL MODEL



- Star schema





FROM FACT TO START SCHEMA

- The basic rule for translating a fact schema into a star schema is to:

Create a fact table containing all the measures and descriptive attributes directly related to the fact and, for each hierarchy, create a dimension table containing all its attributes.

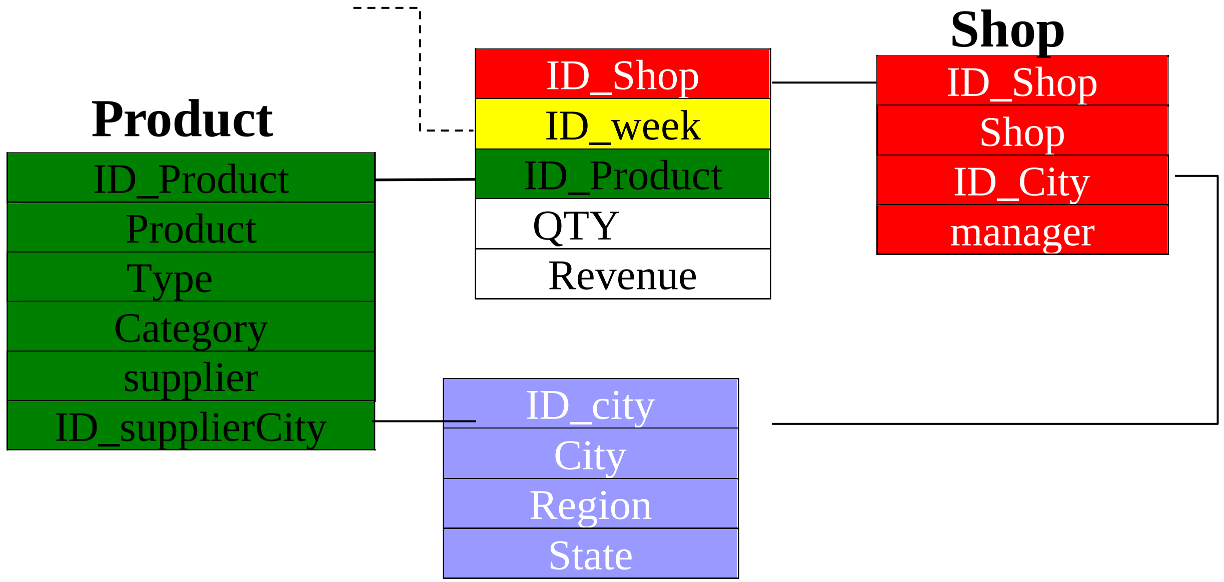
- In addition to this simple rule, the correct translation of a fact schema requires a thorough discussion of advanced DFM constructs.

STAR VS SNOWFLAKE

- There are conflicting opinions on the usefulness of snowflaking:
  - Contrast with the data warehousing philosophy
  - It represents an unnecessary "beautification" of the scheme
  - Inadequate or otherwise more complex to support Type II and III Dynamic Dimensions
- It may be of some use
  - When the ratio between the cardinalities of the primary and secondary dimension table is high, as it determines a (strong) saving of space
  - However, the disk space saving must be evaluated taking into account the performance impact due to the possible additional cost (and complication) of the joins

STAR VS SNOWFLAKE

- Snowflake is useful when
  - ✓ A subset of hierarchies is in common with multiple dimensions



SLOW CHANGING DIMENSIONS

SLOW CHANGING DIMENSIONS

- Hierarchies are properly modeled by functional dependencies
- Functional dependencies imply a static view of reality which, on the other hand, can change over time.
  - the boundaries of the districts are altered, the managers of the shops change,
  - product marketing groups are updated, product categories are changed by department

DYNAMIC HIERARCHIES

- the dynamics of hierarchies influence the logical design of the data warehouse
- depending on how this dynamic will be used by queries, in four scenarios:
  - yesterday for today (type 0)
  - today for yesterday (type 1)
  - yesterday and today (type 2)
  - today or yesterday (type 3)

$$\text{Negozio1} \xrightarrow{[x,\ 01]} \text{ResponsabileRossi}$$

$$\text{Negozio1} \xrightarrow{[02,\ x]} \text{ResponsabileBianchi}$$

$$\text{Negozio1} \xrightarrow{[x,\ 02]} \text{DistrettoPiemonteLiguria}$$

$$\text{Negozio1} \xrightarrow{[03,\ x]} \text{DistrettoPiemonte}$$

  - Responsabile == manager
  - Distretto == District
  - Incassi == revenue

YESTERDAY FOR TODAY

- each event is referred to the original

$$\text{Negozio1} \xrightarrow{[x,\ 01]} \text{ResponsabileRossi}$$

$$\text{Negozio1} \xrightarrow{[02,\ x]} \text{ResponsabileBianchi}$$

$$\text{Negozio1} \xrightarrow{[x,\ 02]} \text{DistrettoPiemonteLiguria}$$

$$\text{Negozio1} \xrightarrow{[03,\ x]} \text{DistrettoPiemonte}$$

Negozio1	2000	2001	2002	2003	2004
Responsabile	Rossi	Rossi	- Bianchi-	Bianchi	Bianchi
Distretto	PiemonteLiguria	PiemonteLiguria	PiemonteLiguria	Piemonte	Piemonte
Incassi	1000	1300	1200	1500	1400

TODAY FOR YESTERDAY

- Events ascribed to the situation of the hier

$$\text{Negozio1} \xrightarrow{[x,\ 01]} \text{ResponsabileRossi}$$

$$\text{Negozio1} \xrightarrow{[02,\ x]} \text{ResponsabileBianchi}$$

$$\text{Negozio1} \xrightarrow{[x,\ 02]} \text{DistrettoPiemonteLiguria}$$

$$\text{Negozio1} \xrightarrow{[03,\ x]} \text{DistrettoPiemonte}$$

Negozio1	2000	2001	2002	2003	2004
Responsabile	Bianchi	Bianchi	Bianchi	Bianchi	Bianchi
Distretto	Piemonte	Piemonte	Piemonte	Piemonte	Piemonte
Incassi	1000	1300	1200	1500	1400

TODAY AND YESTERDAY

- Only events that refer to unchanged values in the hierarchy are considered

TODAY OR YESTERDAY

- Each event refers to the value of the hierarchies in the instant of time in which it occurred

$$\text{Negozio1} \xrightarrow{[x,\ 01]} \text{ResponsabileRossi}$$

$$\text{Negozio1} \xrightarrow{[02,\ x]} \text{ResponsabileBianchi}$$

$$\text{Negozio1} \xrightarrow{[x,\ 02]} \text{DistrettoPiemonteLiguria}$$

$$\text{Negozio1} \xrightarrow{[03,\ x]} \text{DistrettoPiemonte}$$

Negozio1	2000	2001	2002	2003	2004
Responsabile	Rossi	Rossi	- Bianchi-	Bianchi	Bianchi
Distretto	PiemonteLiguria	PiemonteLiguria	PiemonteLiguria	Piemonte	Piemonte
Incassi	1000	1300	1200	1500	1400